

PubMedCausal: A Span-Level Annotated Corpus for Causal Relation Extraction in Biomedical Text

Ifeoluwa Kunle-John^{*1,2} Josiah Paul^{*1,2} Oluwatosin Agbaakin^{†3}

Peter Aina^{‡3} Ikenna Odezuligbo^{†4} Sydney Anuyah^{‡1,3}

Edyah Limited¹, University of Ibadan, NG²

Indiana University, Indianapolis, IN³, Creighton University, Omaha NE⁴

{ikunlejohn, pjosiah, sanuyah}@edyahlimited.com.ng, {ikunle-john252399, pjosiah539}@stu.ui.edu.ng
oluwatosin.agbaakin@alumni.iu.edu, petaina@iu.edu, ikennaodezuligbo@creighton.edu, sanuyah@iu.edu

Abstract

Causal relation extraction (CRE) is central to biomedical text mining, but current resources often conflate causal relations with broader associations, restrict annotation to sentence-level examples, or focus mainly on explicit causal cues. This limits their usefulness for evaluating whether models can recover causal claims as they are actually expressed in biomedical text. We introduce PUBMEDCAUSAL, a span-level annotated corpus for biomedical CRE built from PubMed abstracts. The corpus contains 30,000 paragraph-level rows, including 3,945 causal rows and 6,491 adjudicated cause–effect pairs. Each causal relation is annotated with full-text cause and effect spans, causality type, and sententiality, enabling evaluation of both causal detection and full-span causal extraction. We benchmark discriminative encoders and open-source generative models across detection and extraction settings. For causal detection, biomedical encoders are strongest, with PubMedBERT reaching an F_1 score of 0.7391. For span-level extraction, the best generative baseline is DeepSeek-R1-32B with few-shot prompting, reaching a Cosine Pair F_1 of 0.6765. We further test transfer learning by evaluating PubMedCausal-trained encoders on external causal relation datasets, showing that the resource supports cross-dataset evaluation. Our results show that biomedical CRE remains difficult under class imbalance, long causal spans, implicit causality, inter-sentential relations, and prompt sensitivity. [Code and Data can be found here](#)

1 Introduction

Large Language Models (LLMs) demonstrate strong general language capabilities, yet they continue to struggle with causal extraction (CE), of-

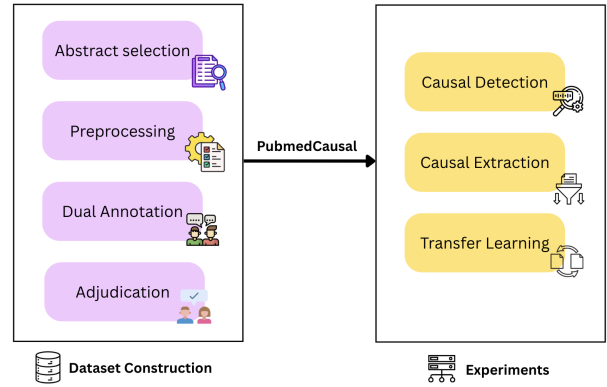


Figure 1: Overview of our work

ten conflating statistical associations with genuine causal relationships (Bazgir et al., 2025). This limitation is especially pronounced for implicit causal relations and for causal relations that extend across sentence boundaries (Anuyah et al., 2025; Zadrozny, 2025). Identifying causality in text requires attention to lexical choice, syntactic composition, discourse relations, and modality, since each can substantially alter the meaning, direction, and strength of a causal claim.

As these linguistic mechanisms recur across domains, they may support a more transferable basis for modeling causality than approaches tied primarily to domain-specific ontologies or knowledge bases (Feder et al., 2022; Singhal et al., 2023; Chen et al., 2025; Lai et al., 2023). We use biomedical literature because it includes many causal relations grounded in experimental and clinical evidence. Reliable CE in biomedical contexts can support downstream applications such as drug discovery, disease management, clinical decision-making, and precision medicine (Prosperi et al., 2020). However, the abundance of biomedical data presents a growing scale problem for manual annotation; as PubMed alone indexes more than 5,000 new articles per day and contains over 37.6 million

^{*}Equal contribution. Co-first authors.

[†]Equal contribution. Co-second authors.

[‡]Project lead.

Table 1: Coverage matrix. ✓ explicit; ∼ implied; × none. **Pair** = structured cause–effect pair annotation; **Exp** = explicit causality; **Imp** = implicit causality; **Intra** = intra-sentential relations; **Inter** = inter-sentential relations; **Span** = span-level (non-atomic) arguments.

Dataset	Domain	Size & Key Features	Pair	Exp	Imp	Intra	Inter	Span
AltLex (Hidey and McKeown, 2016)	General / News	1,027 sents.; 456 causal / 571 non-causal; causal connective labels	×	∼	∼	✓	∼	×
SemEval 2010 Task 8 (Hendrickx et al., 2010)	General / News	10.7K sents.; 9 relation types; 1,003 cause–effect pairs	✓	✓	×	✓	×	×
Causal News Corpus (Tan et al., 2022)	News	3,559 event sents.; binary causal labels; protest-domain sources	×	✓	∼	✓	∼	×
BioCause (Mihăilă et al., 2013)	Biomedical	851 causal relations; 19 full-text articles; infectious disease subdomain	✓	✓	×	✓	✓	∼
FinCausal (Mariko et al., 2020)	Financial	29K segments; 3,405 docs; 4,029 cause tags; 5,350 quantitative fact tags	✓	✓	∼	✓	×	✓
Causal TimeBank (Mirza et al., 2014)	General / News	6,811 events; 318 explicit causal links with source→target annotation	✓	✓	×	✓	✓	×
PubMedCausal (Ours)	Biomedical	30K paragraph rows (PubMed); 3,945 causal rows; 6,491 cause–effect pairs	✓	✓	✓	✓	✓	✓

records (Sayers et al., 2022, 2025). Beyond the published literature, hospitals generate large volumes of electronic health records (EHRs) that may contain clinically relevant causal information, further motivating the need for robust CE methods.

Current automated systems remain limited in their ability to capture this complexity. We did not design a new model to resolve this challenge. Instead, we provide a resource for evaluating CE, enabling deeper examination of causality and uncertainty types while offering a baseline for future work.

Large-scale resources such as CauseNet and Webis Medical CauseNet employ rule-based extraction strategies to collect millions of cause-effect pairs (Heindorf et al., 2020), but these methods remain constrained by precision and by the rigidity of predefined extraction patterns. Existing supervised approaches similarly face important limitations. Relation extraction (RE) datasets frequently conflate causal and associative relationships, while causality-focused datasets are often domain-narrow, limited in scale, or restricted to sentence-level annotations (Bansal et al., 2025; Moghimifar et al., 2020). In addition, these datasets provide limited coverage of implicit or inter-sentential causality, in favour of explicit causality. Synthetic datasets derived from structural causal models (SCMs), such as COPA (Roemmele et al., 2011), offer valuable insights into implicit causality but may introduce statistical patterns that learning algorithms exploit, limiting generalization to naturally occurring text (Ormaniec et al., 2025). Consequently, there remains a need for large-scale, human-annotated resources that preserve the linguistic structure of causal text.

To address these limitations, we propose an annotation framework for CE centered on textual structure and causal expression, and introduce its first large-scale instantiation, PubMedCausal.

We investigate the following research questions:

- **RQ1:** Can current models accurately detect causal spans in text when causal instances are substantially outnumbered by non-causal instances?
- **RQ2:** How effectively can current models extract full-span cause-effect relations from real-world biomedical paragraphs?
- **RQ3:** How does fine-tuning on annotations that preserve textual structure affect extraction performance and robustness across prompting settings?
- **RQ4:** How do models perform on explicit versus implicit causality, and intra-sentential versus inter-sentential relations?

Our main contributions are as follows:

- We present PubMedCausal, a rigorously annotated resource containing 30,000 rows of 2–5 sentences from PubMed abstracts and 6,491 cause-effect pairs across 3,945 causal instances.
- We annotate cause and effect as full-text spans, preserving contextual information relevant to causal interpretation.
- We explicitly distinguish causal and non-causal rows, supporting both causal detection and causal extraction.

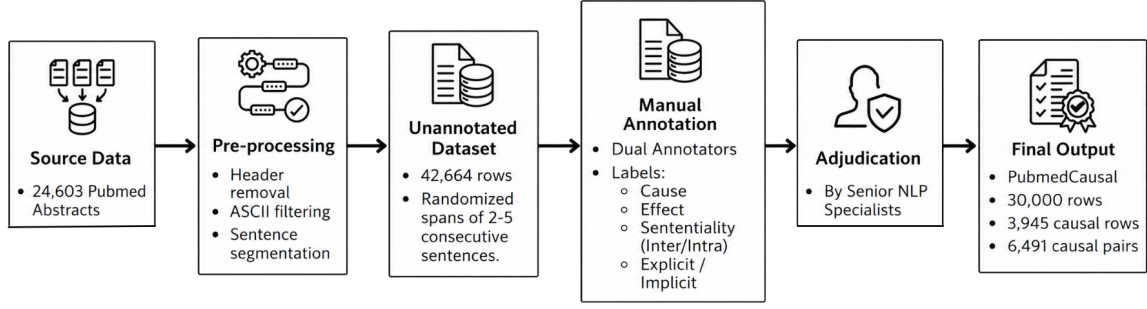


Figure 2: PubMedCausal construction process. Abstracts were retrieved from PubMed, preprocessed, filtered and dual annotated. The final corpus contains 6,491 adjudicated causal relations labeled by sententiality and causality type.

- We annotate causal relations by causality type and sententiality, enabling targeted evaluation of implicit and cross-sentence causality.

2 Related Work

The motivation of this work is to address the lack of human-annotated resources that distinguish causal relations from broader biomedical associations. Existing biomedical RE datasets such as BC5CDR, GAD, DisGeNET, and BioRED have enabled substantial progress in extracting relations among chemicals, genes, diseases, and other biomedical entities (Li et al., 2016; Bravo et al., 2015; Piñero et al., 2016; Luo et al., 2022). However, these resources generally target associative or entity-level relations rather than causal relations, making them insufficient for evaluating whether models can identify causality as expressed in biomedical text.

CE has been studied in general-domain datasets such as AltLex, Causal TimeBank, and the Causal News Corpus (Hidey and McKeown, 2016; Mirza et al., 2014; Tan et al., 2022). These resources capture important forms of causal language, but they are either restricted largely to sentence-level relations, or focused on explicit causal cues. Biomedical causal resources are more limited. BioCause annotates causal relations in biomedical articles (Mihăilă et al., 2013), MIMICause studies causal relation types in clinical notes (Khetan et al., 2022), and CRED distinguishes causal from non-causal gene–disease relations in PubMed abstracts (Bansal et al., 2025). While valuable, these datasets remain limited in scale, domain coverage, or task formulation, particularly for full-span extraction of implicit and cross-sentence causal relations.

Although CauseNet and Webis Medical CauseNet provide broad, automatically extracted

causal knowledge, their usefulness for fine-tuning LLMs may be constrained by their relatively predictable patterns, since they were created by rule-based algorithms and they lack non-causal instances (Heindorf et al., 2020; Schlatt et al., 2022). The presence of only one class makes it less useful for causal detection, and the rule-based patterns can reduce their generalization to real-world contexts.

PubMedCausal addresses these gaps by providing a large-scale, human-annotated biomedical CE resource built from PubMed abstracts. It goes beyond prior biomedical RE datasets by distinguishing causal from non-causal text, and extends existing biomedical CE resources by annotating complete cause and effect spans, covering both explicit and implicit causality, and including intra-sentential as well as inter-sentential relations. Table 1 summarizes these differences.

3 Dataset Construction

3.1 Source Data

PubMedCausal was constructed from PubMed abstracts retrieved using the keyword “causality” from January 1st to September 3rd, 2025. This query returned 24,603 free full-text abstracts, which were segmented into sentences and then combined into rows with 2–5 successive sentences to support the annotation of both intra-sentential and inter-sentential causal relations. To improve readability and reduce annotation complexity, rows containing numbers or non-ASCII characters were removed, leaving 42,664 candidate rows. Instances without causal relations were retained, since the corpus is intended to support both CE and causal/non-causal classification.

Due to annotation cost and time constraints, we annotated 30,000 rows from this candidate pool.

Table 2: Representative annotated examples from PubMedCausal.

Span	Cause	Effect	Sent.	Marker/Causality Type
A causal association between media reporting of suicides and subsequent actual suicides has been observed. This study examines whether educating media professionals about responsible reporting can change the quality of reporting.	media reporting of suicides	subsequent actual suicides	Intra	Explicit
This study found that sexual victimization significantly increased the likelihood of a diagnosis of psychosis and therefore suggests that there may be a role for traumatic experiences in the etiology of psychosis.	sexual victimization	a significant increase in the likelihood of a diagnosis of psychosis	Intra	Implicit
HyCoSy is a well-tolerated examination and the associated vaginal effects are unusual and generally mild. Consequently, we support its introduction as a first-line procedure for tubal patency and uterine cavity investigation in infertile women.	HyCoSy is a well-tolerated examination and the associated vaginal effects are unusual and generally mild	we support its introduction as a first-line procedure for tubal patency [Pair 1]	Inter	Implicit
[same source as above]	HyCoSy is a well-tolerated examination and the associated vaginal effects are unusual and generally mild	we support its introduction as a first-line procedure for uterine cavity investigation in infertile women [Pair 2]	Inter	Implicit
Sepsis is a common condition encountered in hospital environments. There is no effective treatment for sepsis, and it remains an important cause of death at intensive care units. This study aimed to discuss some methods that are available in clinics, and tests that have been recently developed for the diagnosis of sepsis.	Sepsis	death at intensive care units	Inter	Explicit

Further details on the sampling strategy, preprocessing decisions, and corpus scope are provided in Appendix C.1.

3.2 Annotation Protocol

Annotators We called for applications and offered a test to applicants. The top 10 performers were contacted and trained before commencing the annotation task. All annotators followed the same guideline. The remaining rows were split into batches, and two annotators worked on each batch independently.

Annotation Scheme We annotated the text by marking the full cause span and the full effect span for each causal relation. A cause or effect could be a single entity, a noun phrase, or a full clause, depending on how the relation was expressed in the text. If one cause led to more than one separate effect, each cause–effect pair was recorded as a separate row. This applied whether the effects appeared in the same sentence or across different sentences. Where possible, the shared cause span was kept exactly as written in the text, so that separate causal relations were not merged into one broad annotation. Full annotation guidelines and worked examples are provided in Appendix D.2. Each identified relation was annotated along two dimensions:

1. **Sententiality.** *Intra-sentential*: both cause and effect spans occur within the same sentence. *Inter-sentential*: the spans cross more

than one sentence.

2. **Causality Type.** *Explicit*: the causal relation is directly signaled by a lexical causal marker (see Appendix for the full inventory). *Implicit*: the causal interpretation arises from semantic composition, event structure, or discourse context rather than from a dedicated causal expression.

Table 2 presents representative annotated examples from the corpus, illustrating multi-pair annotation, intra-sentential explicit, intra-sentential implicit, and inter-sentential relations.

3.3 Adjudication and Inter-Annotator Agreement

Table 3 summarises pre-adjudication agreement across the main annotation subtasks.

To account for boundary variation in extracted spans, we computed token-level partial match F_1 . For two spans a and b with token sets T_a and T_b , we define:

$$F_{1,\text{tok}}(a, b) = \frac{2|T_a \cap T_b|}{|T_a| + |T_b|}. \quad (1)$$

For texts containing multiple causal relations, annotator outputs were aligned using bipartite matching. Given two relations $r_1 = (C_1, E_1)$ and $r_2 = (C_2, E_2)$, their similarity was:

$$S(r_1, r_2) = \frac{1}{2}F_{1,\text{tok}}(C_1, C_2) + \frac{1}{2}F_{1,\text{tok}}(E_1, E_2). \quad (2)$$

The Hungarian algorithm was then used to find the optimal one-to-one matching under cost

Table 3: Pre-adjudication inter-annotator agreement across annotation subtasks.

Annotation subtask	Metric	Score
Causal span detection	Cohen’s κ	0.48
Causal span detection	Raw agreement	0.90
Cause span extraction	Partial match F_1	0.92
Effect span extraction	Partial match F_1	0.85
Sententiality classification	Cohen’s κ	0.55
Sententiality classification	Raw agreement	0.98
Causality expression type	Cohen’s κ	0.61
Causality expression type	Raw agreement	0.82

$1 - S(r_1, r_2)$. Cause and effect agreement were reported as the mean token-level F_1 over matched relation pairs.

Sententiality classification achieved $\kappa = 0.55$ with raw agreement of 0.98, again reflecting the effect of label imbalance, since most relations were intra-sentential. Causality expression type achieved $\kappa = 0.61$ with raw agreement of 0.82, suggesting moderate agreement and confirming the relative difficulty of distinguishing implicit causal relations from explicit ones.

After annotation, an adjudication panel reviewed all disputed cases. They corrected clear errors, adjusted span boundaries when the original span did not fully capture the causal meaning, and reached a final decision on difficult cases.

3.4 Corpus Statistics

As shown in Figure 3, 26,055 of the 30,000 processed instances (86.85%) contain no annotated causal relation. The remaining 3,945 causal instances yielded 6,491 cause–effect relations, with 2,412 instances containing one pair and 1,533 containing multiple pairs.

This imbalance reflects the scope of our causality-query-derived corpus, not causal prevalence in PubMed overall. Even with a causality-oriented query, most processed instances lack annotatable cause–effect relations. Sampling bias and corpus scope are discussed in Appendix C.2.

4 Experimental Setup

4.1 Model Selection

We evaluate two model families, encoder-based classifiers and open source LLMs for the causal detection task. The encoder models used are BERT, BioBERT, PubMedBERT, SciBERT. For the LLMs, we selected several models across different ranges of parameters from 3B to 70B. Larger models (Llama 70B, DeepSeek-R1 32B, DeepSeek 70B,

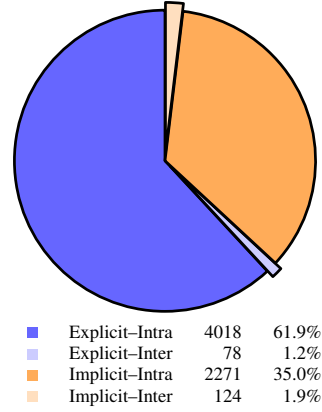


Figure 3: Distribution of causal relations by causality type and sententiality.

Mixtral 8x7B) were evaluated in zero-shot settings under 4-bit quantization, while smaller models (Llama 3.2-3B, Llama 3.1-8B, Qwen 2.5-7B, Mistral-7B, DeepSeek-V3-3B, DeepSeek-V2.5-7B) were not quantized. In addition, we fine-tuned these smaller LLM classes using LoRA.

On the second task, CE, we evaluated only the LLMs, still holding the same configuration for the first task.

4.2 Data Splits

For causal detection, the full 30,000-row corpus was split 50:50 into training and test sets using stratified sampling to maintain the empirical class distribution in both partitions. For causal *extraction*, we focused on the 3,945 causal rows and applied a complexity-stratified split. The causal data was first divided into two clusters: single-pair rows (approximately 61%) and multi-pair rows (approximately 39%). Each cluster was independently split 50:50; the final training and test sets were formed by merging the corresponding halves.

4.3 Prompting Strategies and Evaluation Metrics

Six prompting strategies were tested for all generative models: zero-shot (ZS), few-shot (FS), chain-of-thought (COT), CoT + FS, ReAct, and least-to-most (L2M) prompting. All prompts are provided in the technical appendix E.

For causal detection, we report Precision, Recall and F_1 on the causal class. For extraction, we use a three-tier evaluation: Exact Match F_1 (strict string match), Token Overlap F_1 (partial credit for shared tokens), and Cosine Similarity. Cosine Similarity F_1 is computed using the embedding model from Deka et al. (2022), fine-tuned on MedNLI and

SciNLI, providing domain-appropriate sentence embeddings for biomedical span comparison.

We apply a pre-specified cosine similarity threshold of 0.75, following recent medical literature mining work (Wang et al., 2025). This threshold accommodates minor boundary variation in longer biomedical causal spans while still requiring substantial semantic agreement between predicted and gold spans.

5 Baseline Experiments

We report baseline experiments for the two tasks supported by the corpus. These experiments provide initial points of comparison for future work and illustrate the types of challenges introduced by the dataset, including class imbalance, long causal arguments, implicit causality type, and multi-pair spans.

For each generative model, all six prompting strategies were evaluated and the best performing configuration per model is reported in this section; complete results are provided in Appendix F, informing our choice of the best prompt configuration.

5.1 Experiment 1: Causal Span Detection Baselines

Table 4 reports causal span detection results on PubMedCausal corpus. This task evaluates whether a model can distinguish paragraphs that contain at least one causal relation from non-causal ones. The encoder models provide the strongest reference baseline, especially PubMedBERT, which obtained the highest F_1 score of 0.7391, followed closely by BioBERT with an F_1 score of 0.7380. These results suggest that biomedical pretraining is useful for identifying causal language in PubMed-derived text.

LLMs perform substantially lower as the best prompt-only generative model (Meta-Llama-3.3-70B) had a F_1 score of 0.4086, while the strongest fine-tuned generative model (Qwen-7B-FT) had a F_1 score of 0.3782.

5.2 Experiment 2: Span-Level Causal Pair Extraction Baselines

In experiment 2, we extract the cause and effect pairs. In the annotation, we have observed that both cause and effect can be a singular token or an entire phrase or sentence. This is the motivation for span level extraction which covers all the

Table 4: Reference baseline results for causal span detection on PubMedCausal. Scores are reported for the causal class. Full prompting results are provided in Table 12 Appendix F.1

Model	Prec.	Rec.	F_1
<i>Discriminative Encoders</i>			
BERT	0.7249	0.7344	0.7296
SciBERT	0.7147	0.7440	0.7291
PubMedBERT	0.7289	0.7496	0.7391
BioBERT	0.7282	0.7481	0.7380
<i>Generative Decoders — Prompt-Only (Best Configuration)</i>			
Meta-Llama-3.3-70B (ZS)	0.3004	0.6386	0.4086
DeepSeek-R1-32B (L2M)	0.2264	0.8692	0.3592
Mixtral-8x7B (CoT)	0.2266	0.7481	0.3478
Llama-8B (CoT-FS)	0.2421	0.5666	0.3393
Qwen-7B (CoT)	0.2122	0.5028	0.2984
Mistral-7B (CoT)	0.1782	0.8697	0.2958
Llama-3B (CoT-FS)	0.1825	0.6128	0.2812
DeepSeek-7B (ZS)	0.1597	0.8834	0.2705
<i>Generative Decoders — Fine-Tuned (Best Configuration)</i>			
Qwen-7B-FT (CoT)	0.4649	0.3188	0.3782
Llama-3B-FT (CoT-FS)	0.2094	0.5692	0.3062
Mistral-7B-FT (CoT-FS)	0.2095	0.5682	0.3062
Llama-8B-FT (FS)	0.1759	0.2899	0.2190
DeepSeek-7B-FT (L2M)	0.1350	0.1825	0.1552

ZS = Zero-Shot; FS = Few-Shot; CoT-FS = CoT Few-Shot; L2M = Least-to-Most.

scenarios. Table 5 reports reference baselines for span-level cause-effect pair extraction on causal rows. The task requires the model to identify the full textual cause and effect spans and preserve their directionality. The task is therefore sensitive to boundary errors, missing pairs, duplicated pairs, and cause-effect reversal.

Table 5: For every model the best prompt strategy was reported, full prompt strategy results for all models are provided in Table 15 (Appendix F.4)

Model	Token Overlap P-F ₁	Exact P-F ₁	Cos. P-F ₁
<i>Generative Decoders — Prompt-Only (Best Configuration)</i>			
DeepSeek-R1-32B (FS)	0.5758	0.2383	0.6765
Mistral-7B (L2M)	0.4739	0.1527	0.5958
Mixtral-8x7B (FS)	0.3704	0.1469	0.4390
Qwen-7B (FS)	0.3640	0.1400	0.4283
Llama-8B (ReAct)	0.2764	0.1089	0.3191
Meta-Llama-3.3-70B (ZS)	0.2384	0.1013	0.2676
Llama-3B (ReAct)	0.2059	0.0717	0.2554
<i>Generative Decoders — Fine-Tuned (Best Configuration)</i>			
Mistral-7B-FT (L2M)	0.4245	0.1022	0.5812
Qwen-7B-FT (ZS)	0.3947	0.1293	0.4817
Llama-8B-FT (FS)	0.3175	0.1039	0.3890
Llama-3B-FT (FS)	0.1615	0.0674	0.1919

ZS = Zero-Shot; FS = Few-Shot; L2M = Least-to-Most; P = Pair

The strongest prompt-only baseline is DeepSeek-R1-32B under few-shot prompting. Nevertheless, the observed F_1 scores suggest that automated causal extraction with LLMs remains far from solved, as most models perform below the 50% threshold. These results also indicate that model size alone does not guarantee superior performance, since smaller models can outperform larger models in certain experimental settings.

Among smaller models, Mistral-7B provides the

strongest base-model result under least-to-most prompting. Fine-tuned models do not consistently outperform their base counterparts: Mistral-7B drops after fine-tuning, while Qwen-7B shows marginal Token Overlap improvement but a decline on Exact Pair F_1 , suggesting LoRA adaptation shifts rather than reliably improves extraction performance.

5.3 Experiment 3: Transfer Learning

This experiment was done to evaluate encoder models trained on PubMedCausal on selected external causal relation datasets Table 6. This is to examine whether the models trained on the corpus are capable of transferred learning. Results show stronger transfer to biomedical and general causal datasets than to financial causal text, suggesting that domain and annotation-schema differences remain important factors in cross-dataset evaluation. Full cross-dataset training parameters and dataset configurations are provided in Appendix G Table 11.

Table 6: Diagnostic cross-dataset causal detection results. PubMedCausal-trained models are evaluated on external datasets to estimate transfer behaviour across domains and annotation schemes.

Dataset	Model	Within F_1	Cross F_1	Δ
FinCausal	BERT	0.2113	0.2222	+0.0109
	SciBERT	0.5333	0.2759	-0.2574
	PubMedBERT	0.5098	0.3448	-0.1650
	BioBERT	0.2800	0.2128	-0.0672
CTB	BERT	0.2009	0.2418	+0.0409
	SciBERT	0.2430	0.1810	-0.0620
	PubMedBERT	0.2181	0.2015	-0.0166
	BioBERT	0.2129	0.1786	-0.0343
BioCause	BERT	0.3529	0.5556	+0.2027
	SciBERT	0.4000	0.5333	+0.1333
	PubMedBERT	0.4478	0.5517	+0.1039
	BioBERT	0.5500	0.5000	-0.0500
AltLex	BERT	0.6696	0.7525	+0.0829
	SciBERT	0.6332	0.7150	+0.0818
	PubMedBERT	0.6723	0.7182	+0.0459
	BioBERT	0.6691	0.7447	+0.0756

CTB = Penn Discourse Treebank-style causal text bank dataset. Within = model fine-tuned and evaluated on the same dataset. Cross = model trained on PubMedCausal and evaluated zero-shot on the target dataset. Change = Cross F_1 - Within F_1 .

6 Analysis and Ablation Studies

6.1 Error Analysis

We conduct error analysis using the strongest extraction baseline, DeepSeek-R1-32B with few-shot prompting. Figure 4 groups predictions into error buckets indicated with a red bar. The dominant error was where the model recovers either the cause or the effect but not both accurately. This shows

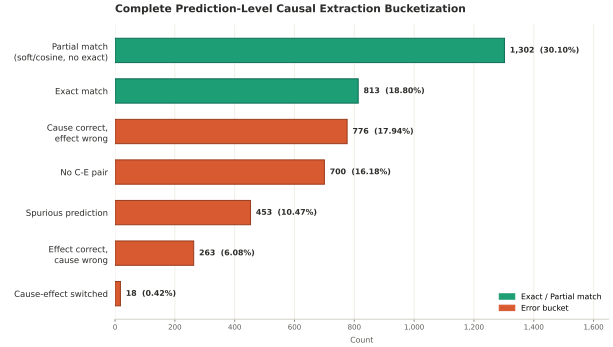


Figure 4: Error bucketization of the model/prompt strategy with the best performance

that PubMedCausal is challenging not only for detecting causal language, but also for recovering complete span-level causal relations. We provide full error bucket definitions and representative examples in Appendix G.2.

6.2 Performance by Causality Type and Sententiality

The experiment treated causality type and sententiality classification as an offshoot of the extraction task. Based on the best extraction model, as shown in Table 7, the model performed better on direct and structurally simpler cases. Inter-sentential causality remains particularly challenging, with a baseline F_1 score of only 5%, indicating substantial room for improvement. In contrast, intra-sentential causality achieved an F_1 score of 97.43%, showing that the model is far more effective when causal relations are expressed within a single sentence than when they must be inferred across sentence boundaries.

In observing the causality type, we see that explicit causality achieved an F_1 score of 88.03%, indicating strong performance when causal relations are clearly marked in the text. In contrast, implicit causality achieved an F_1 score of 39.20%, showing that the model struggles when causality must be inferred.

Model	Explicit F_1	Implicit F_1	Intra F_1	Inter F_1
DS-R1-32B (FS)	0.8803	0.3920	0.9743	0.0500

Table 7: Extraction F_1 broken down by causality type and sententiality. Inter-sentential and implicit causality type are consistently harder across all configurations.

6.3 Model Analysis

LoRA fine-tuning does not consistently improve span-level extraction: Mistral-7B performs better

in its base setting, while Qwen-7B shows marginal gains on Token Overlap F_1 only Table 24.

The effect of causal complexity is model-dependent: Llama-3B and Llama-8B perform worse on multi-pair spans, while Mistral-7B and Qwen-7B perform better on multi-pair spans. This indicates that multi-pair extraction difficulty is not uniform across models, but depends on the interaction between model family and prompting strategy. Full ablation tables are in Appendix H.

7 Discussion

We observe that causal detection is difficult because causal references in the real-world are not mentioned in every sentence, and as such, this huge imbalance has made it difficult for models to accurately pick up on what is causal and what is not. In our causally biased corpus, we observe approximately a 7:43 split between causal and non-causal rows. Therefore, the prompting approach makes models more preemptive to explore causal statements even when none is present. The inclusion of encoder models as a first filter for causal statements could offer more practical benefits than using standalone LLMs. Even when a model correctly discovers that a sentence is causal, the ability to perform extraction requires the model to recover complete causal arguments, including long texts, implicit causality type, and multiple pairs within a single row. From our results, we see that there is still a need to develop frameworks that can handle automated extraction at a higher level of precision and recall.

PubMedCausal can therefore support future work on causal filtering, full-span extraction, implicit causality type and inter-sententiality, and pipeline-based systems that combine discriminative detection with generative extraction (ZAVARELLA et al., 2025).

8 Conclusion

We introduced PubMedCausal, a 30,000-row annotated corpus for CE in biomedical text. The dataset captures span-level causal arguments, full descriptive phrases rather than atomic entities, annotated along dimensions of marker presence and sententiality. Its annotation protocol, grounded in semantics, is designed to prevent models from exploiting pre-trained domain knowledge during evaluation.

Empirical evaluation across encoder classifiers and open-source generative models shows that the

dataset poses genuine difficulty. Encoder models substantially outperform generative models on binary causal detection under realistic class imbalance. Fine-tuning does not produce consistent improvements, as performance varies across prompt strategies and fine-tuned models generally do not surpass the best prompt-only baselines. The fine-tuning setup was held fixed across models to ensure comparability rather than optimized per model. We release all model configurations, prompts, and data splits to enable direct comparison, and encourage future work to build on these baselines with repeated evaluation and broader hyperparameter search.

Limitations

Several limitations should be considered when interpreting PubMedCausal and the reported baselines.

- **Corpus scope.** PubMedCausal is limited to English PubMed abstracts retrieved with the keyword “causality.” This helped us collect more causal language within the annotation budget, but it also means the corpus is not a representative sample of all biomedical writing. It may contain more abstracts on causal inference, epidemiology, and risk-factor analysis, while missing causal statements expressed through other triggers such as “induces,” “reduces,” “contributes to,” or “leads to.” As a result, performance on PubMedCausal may not directly generalize to full-text articles, clinical notes, non-English biomedical text, or other domains.
- **Preprocessing effects.** Some preprocessing choices narrowed the corpus. Rows with numbers or non-ASCII characters were removed to make annotation easier, but this may exclude causal claims involving measurements, statistical results, gene symbols, dosage information, or trial outcomes. Also, the large number of non-causal rows should be seen as a property of this query-derived corpus, not as an estimate of how common causal claims are in PubMed.
- **Class imbalance.** Explicit intra-sentential causal relations are much more common than implicit and inter-sentential relations. This makes implicit and cross-sentence causal

cases harder to evaluate reliably, since fewer examples are available for those subgroups.

- **Annotation subjectivity.** Although the corpus was dual annotated and adjudicated, causal relation extraction still involves judgment. Annotators may differ on the exact boundaries of long cause and effect spans. Implicit causality can also be difficult to separate from association, prediction, or background scientific discussion. PubMedCausal captures causal claims as they are written in the text; it does not verify whether those claims are biologically or clinically true.
- **Baseline scope.** The experiments are intended as reproducible reference baselines, not fully optimized systems. Hyperparameters were kept fixed across models for comparability, and results are based on controlled single-run evaluations without variance across seeds or prompt choices.
- **Metric limitations.** The automatic extraction metrics have known limits. Exact match can penalize reasonable span-boundary differences, while softer metrics may not fully capture causal direction, biomedical specificity, or output-format errors. Future work should evaluate PubMedCausal across broader biomedical subdomains, include repeated runs, expand implicit and inter-sentential cases, and test stronger model- and prompt-specific optimization.

Dataset Release. PubMedCausal will be released via Hugging Face Datasets at [repository link to be added upon acceptance]. The release includes the full 30,000-row corpus in JSON Lines format, with fields for the source PubMed abstract ID, the row text, a binary causal label, and—for causal rows—a list of cause-effect pair tuples, each containing `cause_span`, `effect_span`, `expression_type` (Explicit/Implicit), and sententiality (Intra/Inter). Pre-constructed train and test split indices are provided alongside the data. Raw abstract text is released where permitted under PubMed’s open access policy; for remaining records, PubMed IDs are provided to allow retrieval.

References

Sydney Anuyah, Sneha Shajee-Mohan, Ankit-Singh Chauhan, and Sunandan Chakraborty. 2025. Bench-

marking llms for pairwise causal discovery in biomedical and multi-domain contexts. In *2025 IEEE International Conference on Big Data (Big-Data)*, pages 7953–7962. IEEE.

Nency Bansal, R C Sri Dhinesh, Ayush Pathak, and Manikandan Narayanan. 2025. Beyond associations: A benchmark Causal Relation Extraction Dataset (CRED) of disease-causing genes, its comparative evaluation, interpretation and application. *bioRxiv*.

Adib Bazgir, Amir Habibdoust Lafmajani, and Yuwen Zhang. 2025. Beyond correlation: Towards causal large language model agents in biomedicine. *arXiv preprint arXiv:2505.16982*.

Àlex Bravo, Janet Piñero, Núria Queralt-Rosinach, Michael Rautschka, and Laura I Furlong. 2015. Extraction of relations between genes and diseases from text and large-scale data analysis: implications for translational research. *BMC bioinformatics*, 16(1):55.

Qingyu Chen, Yan Hu, Xueqing Peng, Qianqian Xie, Qiao Jin, Aidan Gilson, Maxwell B Singer, Xuguang Ai, Po-Ting Lai, Zhizheng Wang, and 1 others. 2025. Benchmarking large language models for biomedical natural language processing applications and recommendations. *Nature communications*, 16(1):3280.

Pritam Deka, Anna Jurek-Loughrey, and 1 others. 2022. Evidence extraction to validate medical claims in fake news detection. In *International Conference on Health Information Science*, pages 3–15. Springer.

Amir Feder, Katherine A Keith, Emaad Manzoor, Reid Pryzant, Dhanya Sridhar, Zach Wood-Doughty, Jacob Eisenstein, Justin Grimmer, Roi Reichart, Margaret E Roberts, and 1 others. 2022. Causal inference in natural language processing: Estimation, prediction, interpretation and beyond. *Transactions of the Association for Computational Linguistics*, 10:1138–1158.

Stefan Heindorf, Yan Scholten, Henning Wachsmuth, Axel-Cyrille Ngonga Ngomo, and Martin Potthast. 2020. Causenet: Towards a causality graph extracted from the web. In *Proceedings of the 29th ACM international conference on information & knowledge management*, pages 3023–3030.

Iris Hendrickx, Su Nam Kim, Zornitsa Kozareva, Preslav Nakov, Diarmuid Ó Séaghdha, Sebastian Padó, Marco Pennacchiotti, Lorenza Romano, and Stan Szpakowicz. 2010. [SemEval-2010 task 8: Multi-way classification of semantic relations between pairs of nominals](#). In *Proceedings of the 5th International Workshop on Semantic Evaluation*, pages 33–38, Uppsala, Sweden. Association for Computational Linguistics.

Christopher Hidey and Kathleen McKeown. 2016. Identifying causal relations using parallel wikipedia articles. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1424–1433.

- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang Wang, Weizhu Chen, and 1 others. 2022. Lora: Low-rank adaptation of large language models. *Iclr*, 1(2):3.
- Vivek Khetan, Md Imbesat Rizvi, Jessica Huber, Paige Bartusiak, Bogdan Sacaleanu, and Andrew Fano. 2022. Mimicause: Representation and automatic extraction of causal relation types from clinical notes. In *Findings of the association for computational linguistics: ACL 2022*, pages 764–773.
- Tuan Manh Lai, ChengXiang Zhai, and Heng Ji. 2023. Keblm: knowledge-enhanced biomedical language models. *Journal of Biomedical Informatics*, 143:104392.
- Jiao Li, Yueping Sun, Robin J Johnson, Daniela Sciaky, Chih-Hsuan Wei, Robert Leaman, Allan Peter Davis, Carolyn J Mattingly, Thomas C Wieggers, and Zhiyong Lu. 2016. Biocreative v cdr task corpus: a resource for chemical disease relation extraction. *Database*, 2016.
- Ling Luo, Nina Poerner, Michael Scerri, and Benjamin Glaser. 2022. BioRED: A rich biomedical relation extraction dataset. *Briefings in Bioinformatics*, 23(5):bbac343.
- Dominique Mariko, Hanna Abi-Akl, Estelle Labidurie, Stephane Durfort, Hugues De Mazancourt, and Mahmoud El-Haj. 2020. The financial document causality detection shared task (FinCausal 2020). In *Proceedings of the 1st Joint Workshop on Financial Narrative Processing and MultiLing Financial Summarisation*, pages 23–32, Barcelona, Spain (Online). COLING.
- Claudiu Mihăilă, Mihaela Oprea, and Sophia Ananiadou. 2013. Biocause: Annotating and analysing causality in the biomedical domain. *BMC Bioinformatics*, 14(1):2.
- Paramita Mirza, Rachele Sprugnoli, Sara Tonelli, and Manuela Speranza. 2014. [Annotating causality in the TempEval-3 corpus](#). In *Proceedings of the EACL 2014 Workshop on Computational Approaches to Causality in Language (CAtoCL)*, pages 10–19, Gothenburg, Sweden. Association for Computational Linguistics.
- Farhad Moghimifar, Gholamreza Haffari, and Mahsa Baktashmotlagh. 2020. Domain adaptive causality encoder. In *Proceedings of the 18th Annual Workshop of the Australasian Language Technology Association*.
- Weronika Ormaniec, Scott Sussex, Lars Lorch, Bernhard Schölkopf, and Andreas Krause. 2025. Standardizing structural causal models. In *The Thirteenth International Conference on Learning Representations (ICLR 2025)*.
- Janet Piñero, Àlex Bravo, Núria Queralt-Rosinach, Alba Gutiérrez-Sacristán, Jordi Deu-Pons, Emilio Centeno, Javier García-García, Ferran Sanz, and Laura I Furlong. 2016. Disgenet: a comprehensive platform integrating information on human disease-associated genes and variants. *Nucleic acids research*, page gkw943.
- Mattia Prosperi, Yi Guo, Matt Sperrin, James S Koopman, Jae S Min, Xing He, Shannan Rich, Mo Wang, Iain E Buchan, and Jiang Bian. 2020. Causal inference and counterfactual prediction in machine learning for actionable healthcare. *Nature Machine Intelligence*, 2(7):369–375.
- Melissa Roemmele, Cosmin Adrian Bejan, and Andrew S Gordon. 2011. Choice of plausible alternatives: An evaluation of commonsense causal reasoning. In *AAAI spring symposium: logical formalizations of commonsense reasoning*, pages 90–95.
- E. W. Sayers, J. Beck, E. E. Bolton, J. R. Brister, J. Chan, R. Connor, M. Feldgarden, A. M. Fine, K. Funk, J. Hoffman, S. Kannan, C. Kelly, W. Klimke, S. Kim, S. Lathrop, A. Marchler-Bauer, T. D. Murphy, C. O’Sullivan, E. Schmieder, and 8 others. 2025. [Database resources of the national center for biotechnology information in 2025](#). *Nucleic Acids Research*, 53(D1):D20–D29.
- Eric W Sayers, Evan E Bolton, J Rodney Brister, Kathi Canese, Jessica Chan, Donald C Comeau, Catherine M Farrell, Michael Feldgarden, Anna M Fine, Kathryn Funk, and 1 others. 2022. Database resources of the national center for biotechnology information in 2023. *Nucleic acids research*, 51(D1):D29.
- Ferdinand Schlatt, Dieter Bettin, Matthias Hagen, Benno Stein, and Martin Potthast. 2022. Mining health-related cause-effect statements with high precision at large scale. In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 1925–1936.
- Karan Singhal, Shekoofeh Azizi, Tao Tu, S Sara Mahdavi, Jason Wei, Hyung Won Chung, Nathan Scales, Ajay Tanwani, Heather Cole-Lewis, Stephen Pfohl, and 1 others. 2023. Large language models encode clinical knowledge. *Nature*, 620(7972):172–180.
- Fiona Anting Tan, Ali Hürriyetoglu, Tommaso Caselli, Nelleke Oostdijk, Tadashi Nomoto, Hansi Het-tiarachchi, Iqra Ameer, Onur Uca, Farhana Ferdousi Liza, and Tiancheng Hu. 2022. The causal news corpus: Annotating causal relations in event sentences from news. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 2298–2310.
- Zifeng Wang, Lang Cao, Qiao Jin, Joey Chan, Nicholas Wan, Behdad Afzali, Hyun-Jin Cho, Chang-In Choi, Mehdi Emamverdi, Manjot K Gill, and 1 others. 2025. A foundation model for human-ai collaboration in medical literature mining. *Nature communications*, 16(1):8361.
- Wlodek W Zadrozny. 2025. Challenges and opportunities in causality analysis using large language models. *Entropy*, 28(1):23.

Vanni ZAVARELLA, Lorenzo BERTOLINI, Sergio
CONSOLI, Gianni FENU, RECUPERO Diego RE-
FORGIATO, Alessandro ZANI, and 1 others. 2025.
Llm-powered knowledge graph of causal relations in
drug reviews. In *CEUR WORKSHOP PROCEED-
INGS*.

Appendix

A PubMedCausal Details

Causal Pairs	Instances	Corpus Share (%)
0	26,055	86.85%
1	2,412	8.04%
2	971	3.24%
3	315	1.05%
≥ 4	247	0.82%

Table 8: Distribution of cause–effect pairs per sentence.

Causality Type	Intra-	Inter-	Total
Explicit	4,018	78	4,096
Implicit	2,271	124	2,395
Total	6,289	202	6,491

Table 9: Distribution of extracted causal relations by sententiality and expression type.

B Fine-Tuning Configuration

Encoder models. All four encoder classifiers (BERT, SciBERT, PubMedBERT, BioBERT) were fine-tuned for binary sequence classification using a learning rate of 2×10^{-5} , a batch size of 16, and 3 epochs. Training used a 50:50 downsampled split to mitigate the natural class imbalance in biomedical corpora, where non-causal sentences substantially outnumber causal ones. The test split was left at its natural distribution to reflect realistic deployment conditions. Full cross-dataset experimental settings are in Table 11.

Generative models. Generative models were adapted with LoRA (Hu et al., 2022) to reduce computational requirements while preserving task-specific extraction behaviour. We used rank $r=8$ targeting only query and value projections, reducing trainable parameters by approximately 60% relative to full LoRA. Restricting adaptation to q_{proj} and v_{proj} is standard practice: these projections govern attention-based retrieval and are the most impactful for adapting to structured output formats without over-fitting. 8-bit quantisation during training enabled larger effective batch sizes within GPU memory constraints, while 4-bit NF4 quantisation was applied at inference for the 32B model only, as smaller models fit in 16-bit precision. Greedy decoding (temperature 0.0) was used throughout to ensure reproducibility across prompt strategies. Key hyperparameters are summarised in Table 10.

Generative Model Implementation Details. All generative models were loaded from their official Hugging Face checkpoints: meta-llama/Llama-3.2-3B-Instruct, meta-llama/Llama-3.1-8B-Instruct, mistralai/Mistral-7B-Instruct-v0.3, Qwen/Qwen2.5-7B-Instruct, deepseek-ai/DeepSeek-R1-Distill-Qwen-32B, and meta-llama/Llama-3.3-70B-Instruct. Maximum input length was set to 2,048 tokens; maximum output tokens were set to 512 for detection and 1,024 for extraction. Outputs were parsed by searching for structured tuple patterns (Cause:, Effect:, Causality_Type:, Sententiality:) using rule-based regex; responses that did not contain at least one parseable tuple were treated as null predictions and counted as false negatives. Inference was run on a single NVIDIA A100 80GB GPU with a batch size of 1 for 32B models and 4 for all smaller models. Quantization used bitsandbytes 4-bit NF4 for the 32B model and FP16 for all others. Few-shot examples in prompts were drawn from the training split only; no test examples were used as in-context demonstrations.

Table 10: Fine-Tuning Configuration for Generative Models.

Parameter	Value
<i>LoRA Configuration</i>	
Rank (r)	8
Alpha (α)	16
Target modules	$q_{\text{proj}}, v_{\text{proj}}$
Dropout	0.05
<i>Training Hyperparameters</i>	
Optimizer	AdamW
Learning rate	2×10^{-4}
LR schedule	Cosine with 50-step warmup
Gradient clipping	1.0
Per-device batch size	2–4 (model-dependent)
Gradient accumulation steps	8
Effective batch size	16–32
Training epochs	3
<i>Quantisation & Precision</i>	
Training quantisation	8-bit with double quantisation
Training precision	FP16
Inference quantisation	4-bit NF4 (32B+ models only)

C Source Data

C.1 Sampling Strategy, Preprocessing, and Corpus Scope

PubMedCausal was constructed from PubMed abstracts retrieved using the keyword “causality” over a defined publication period. This retrieval strategy was used as an enrichment step rather than as a representative sampling procedure. Since annotatable cause–effect relations are relatively sparse

Table 11: Experimental Setup for Cross-Dataset Causal Detection.

Component	Setting
Task	Binary causal detection
Model family	Encoder classifiers
Models	BERT, SciBERT, PubMedBERT, BioBERT
Split	70% train / 30% test
Split strategy	Stratified where possible
Training balance	Downsample train split to 1:1 class balance
Test balance	Unchanged / natural distribution
Validation	Slice from balanced train split
Cross source	PubMedCausal
Within eval.	Model tested on its own dataset test split
Cross eval.	PubMedCausal-trained model on other datasets
Seed	42
Decision threshold	0.5
Max sequence length	256
Epochs	3
Batch size	16 train / 32 eval
Learning rate	2×10^{-5}
Weight decay	0.01
Warmup ratio	0.1
Early stopping	Patience = 2
Best model metric	F1
Precision	FP16 when CUDA is available

in general biomedical abstracts, a purely random PubMed sample would have required a substantially larger annotation budget to obtain enough causal instances for model development and evaluation. The keyword-based retrieval strategy therefore increased the likelihood of collecting abstracts in which causal reasoning, causal inference, or causality-related discussion might occur.

The initial PubMed search returned 24,603 abstracts. These abstracts were first segmented into sentences. Since the corpus is designed to capture both intra-sentential and inter-sentential causality, neighbouring sentences were then combined into 2–5 sentence spans. This span-based design allows the annotation to preserve local discourse context, especially where a cause is introduced in one sentence and its effect is stated or implied in a following sentence.

After span construction, we removed spans containing numbers or non-ASCII characters to improve readability and reduce annotation difficulty. This preprocessing step left 42,664 candidate spans. We initially planned to annotate 40,000 spans; however, because of annotation cost and time constraints, the final annotated corpus contains 30,000 instances. Each instance therefore consists of either a single sentence or a short paragraph-like span.

Rows containing no causal relation were retained in the corpus. This decision reflects the intended modelling setting: systems should not assume that every biomedical text span contains a cause–effect relation. Instead, they must first distinguish causal from non-causal discourse before extracting causal

pairs. Retaining non-causal instances also supports causal/non-causal classification as a baseline task and makes the extraction setting more realistic than a corpus containing only positive causal examples.

C.2 Sampling Bias and Corpus Scope

The PubMedCausal corpus should be interpreted as a causality-query-derived biomedical corpus rather than as a representative sample of PubMed biomedical writing. The source abstracts were retrieved from PubMed using the keyword “causality” over a defined time period. This retrieval strategy was chosen because the project aimed to construct a biomedical causality resource under practical annotation constraints. Since annotatable causal relations are relatively sparse in scientific abstracts, querying for causality-related papers increased the likelihood of obtaining documents in which causal reasoning, causal inference, or causal discussion might occur.

This sampling strategy introduces an important scope limitation. Abstracts retrieved using the word “causality” are likely to overrepresent papers concerned with causal inference, epidemiology, observational studies, methodology, risk-factor analysis, and related forms of biomedical reasoning. Conversely, the corpus may underrepresent ordinary biomedical causal statements that do not explicitly use the word “causality” at the abstract or metadata level. Many biomedical abstracts express causal content using verbs and constructions such as “induces,” “increases,” “reduces,” “leads to,” “is associated with,” or “contributes to,” without using the term “causality.” PubMedCausal should therefore not be treated as a random or fully representative PubMed corpus.

This limitation also affects how the non-causal rate should be interpreted. In the processed corpus, 26,055 of 30,000 instances contain no annotated causal relation, corresponding to 86.85% of the data. This figure should not be read as an estimate of the prevalence of causal and non-causal sentences across PubMed as a whole. Rather, it shows that even within a corpus intentionally retrieved using a causality-oriented query, many sentences or short textual units remain descriptive, methodological, contextual, associative, or background-oriented rather than containing annotatable cause–effect relations. The high non-causal proportion is therefore informative for modelling within this query-derived setting, but it is not presented as a general statistical claim about biomedical scientific writing.

Despite this sampling bias, the corpus remains useful as a focused resource for biomedical causal relation detection and extraction. First, it provides a testbed for identifying causal relations in abstracts where causal reasoning is likely to be relevant but is not always explicitly realised at the sentence level. Second, retaining non-causal rows makes the dataset suitable for realistic causal relation detection, since models must distinguish true causal statements from surrounding scientific discourse rather than assuming that every instance contains a causal pair. Third, the corpus captures a difficult annotation setting in which causal claims are mixed with associative, evidential, and methodological language, a common feature of biomedical writing.

Future extensions of PubMedCausal could reduce this sampling limitation by adding randomly sampled PubMed abstracts, MeSH-stratified samples, disease-area-specific subsets, and retrieval queries based on broader causal trigger terms such as “induces,” “causes,” “increases,” “reduces,” “leads to,” and “associated with.” Such extensions would make it possible to compare causal language across query-derived, disease-specific, and randomly sampled biomedical corpora.

A further preprocessing limitation is that sentences containing numbers or non-ASCII characters were removed to improve annotation readability; this may exclude some quantitative biomedical causal statements involving measurements, trial outcomes, gene symbols, or statistical results.

D Annotation

D.1 Annotation Protocol

The annotation task required annotators to identify directed causal relations in biomedical text. A causal relation was annotated only when the passage expressed that one event, condition, process, exposure, intervention, or state brought about, contributed to, increased, decreased, prevented, enabled, or otherwise affected another event, condition, process, or outcome. Each valid annotation therefore required a complete cause–effect pair. Passages that contained only correlation, co-occurrence, comparison, prediction, temporal succession, or general association were not annotated as causal unless the text explicitly or strongly implied a directed causal mechanism.

Annotators were instructed to read each row as a single local discourse unit. A row could contain one sentence or a short paragraph, preserving the

immediate context in which causal claims appear in biomedical abstracts. If both the cause and effect occurred within the same sentence, the relation was labelled as *intra-sentential*. If the cause appeared in one sentence and the effect appeared in another sentence within the same row, the relation was labelled as *inter-sentential*. Relations were excluded if either argument was unavailable or could not be clearly identified from the row.

Causal arguments were expected to be explicit, meaningful spans. Pronouns, vague references, and allusions were not annotated as cause or effect spans unless their antecedents were available within the same row. For example, a sentence such as *It caused hunger* was excluded if *it* could not be resolved locally. However, if the previous sentence identified the antecedent, annotators resolved the pronoun and used the explicit antecedent in the cause–effect pair. Annotators were also instructed not to reject long biomedical spans when the complete cause or effect required a multi-word phrase.

Causal relations were classified as either *explicit* or *implicit*. Explicit causality was identified when the text contained an overt causal connective, causal verb, or causal nominal expression. Examples include *cause*, *because*, *due to*, *owing to*, *lead to*, *result in*, *therefore*, *consequently*, *as a result*, *reason*, *primary reason*, *increase*, *decrease*, *promote*, *reduce*, *prevent*, and related forms. Implicit causality was annotated only when the causal relation was recoverable from the local context without an overt marker. Mere temporal ordering was insufficient unless the passage made the causal interpretation clear.

Annotators were explicitly warned against over-annotation. A sentence containing causal vocabulary was not automatically annotated as causal. Mentions of a possible causal relation, causal association, causal link, or the need to investigate causality were excluded when the passage did not assert a directed cause–effect relation. Reported claims were also excluded when the passage merely attributed a causal claim to another source without presenting it as an asserted finding. Similarly, speculative or hedged expressions such as *may cause*, *might result in*, *could lead to*, or *possibly increases* were excluded when the relation was presented as uncertain.

The modal *can* was treated separately from speculative modals. Annotators included *can* constructions when they expressed a general causal capacity or established potential effect. For example, *un-*

treated hypertension can increase stroke risk was annotated because the statement presents hypertension as having the capacity to increase stroke risk. By contrast, *hypertension may increase stroke risk* was excluded when the wording presented the relation as uncertain.

Negative causal assertions were not annotated as positive cause–effect pairs. For example, *exercise did not cause mortality* was excluded because the sentence denies the causal relation. However, causally directed reduction, prevention, or protective effects were included when the passage asserted an actual effect, as in *exercise reduced mortality risk* or *vaccination prevented severe disease*.

D.2 Annotation Guidelines

The guidelines below were issued to human annotators responsible for building the PubMedCausal gold standard. Annotators were required to pass a qualification round before receiving data batches, and their output was cross-validated to compute inter-rater reliability. The schema (Section 2 of the box) directly defines the four-field tuple used throughout all experiments: {Cause, Effect, Type, Sententiality}.

Annotation Guidelines: Causality in Biomedical Text

1. Overview and Workflow

Annotators were asked to identify directed cause–effect relations in biomedical text. The task focused on relations that are supported by the provided text, rather than by external knowledge or assumptions. Each row was treated as a local discourse unit, and annotators were instructed to extract all valid causal relations expressed within that row.

- **Batch Size:** Data was assigned in fixed-size batches.
- **Submission:** Completed batches were submitted for review before subsequent batches were assigned.
- **Quality Review:** Submitted annotations were checked for missing relations, over-annotation, span-boundary errors, incorrect sententiality labels, and inconsistent interpretation of the guidelines.

2. Extraction Schema

For every identified causal relation, annotators extracted a tuple with four fields:

{Cause, Effect, Type, Sententiality}.

Cause: The event, condition, process, exposure, intervention, or state that brings about or influences another outcome.

Effect: The event, condition, process, outcome, or state that is brought about or influenced by the cause.

Type: Whether the causal relation is expressed explicitly or implicitly.

Sententiality: Whether the cause and effect occur within the same sentence or across sentence boundaries.

3. Inclusion and Exclusion Criteria

- Include a relation only when both the cause and the effect can be clearly identified from the provided row.
- Include causal relations expressed through clear causal predicates or constructions, including cases where the text states that one factor increases, decreases, promotes, reduces, prevents, induces, leads to, results in, or is the reason for another.
- Include *can* constructions only when *can* expresses a general causal capacity or established potential effect, such as “*untreated hypertension can increase stroke risk*.”
- Exclude speculative or uncertain relations when the text presents the causal link as unconfirmed, hypothetical, or requiring further investigation, for example with forms such as *may cause*, *might result in*, *could lead to*, or *possibly increases*.
- Exclude purely associative, correlational, predictive, comparative, or co-occurrence statements unless the text supports a directed causal interpretation.

- Exclude mentions of possible causal links when no direction is asserted, such as “*the causal relationship between X and Y was investigated.*”
- Exclude reported or attributed claims when the passage merely reports that a causal claim was made, rather than presenting the relation as an asserted finding within the text.
- Exclude relations where either the cause or effect is vague, pronominal, or unavailable in the row. Pronouns may be resolved only when their antecedents appear within the same row.
- Annotate strictly from the provided text. Do not rely on external biomedical knowledge, external definitions, or assumptions about abbreviations that are not explained in the row.

4. Labelling Conventions

- **Explicit:** Use when the relation is signalled by an overt causal connective, causal verb, or causal nominal expression. Examples include *because, due to, owing to, as a result, therefore, consequently, leads to, results in, causes, increases, decreases, promotes, reduces, prevents, reason, and primary reason.*
- **Implicit:** Use when the causal relation is present but is not signalled by a dedicated causal marker. The relation must still be recoverable from the wording, event structure, or local discourse context.
- **Intra-sentential:** Use when both the cause and effect occur within the same sentence.
- **Inter-sentential:** Use when the cause and effect occur in different sentences within the same row.

5. Structural Rules

- Extract the complete meaningful span for both cause and effect. A cause or effect may be a single entity, a noun

phrase, or a full clause, depending on what is needed to preserve the meaning of the relation.

- Decompose multiple independent effects into separate cause–effect pairs. For example, “*A causes B and C*” should be annotated as two relations: *A causes B* and *A causes C*.
- If multiple factors jointly produce one effect, keep the combined cause as a single span when the text presents them as jointly necessary or jointly explanatory.
- Do not infer transitive causal closure. If the text states “*A causes B, and B causes C,*” annotate *A causes B* and *B causes C*. Do not annotate *A causes C* unless the text explicitly supports that relation.
- Preserve causal direction regardless of grammatical voice. For passive constructions such as “*Y is caused by X,*” annotate **Cause:** X and **Effect:** Y.
- Do not annotate denied causal relations as positive cause–effect pairs. For example, “*X did not cause Y*” should be excluded. However, asserted reduction or prevention relations should be included, as in “*X reduced Y*” or “*X prevented Y.*”

6. Practical Annotation Checks

- Check whether the extracted pair can be paraphrased as: *Cause brings about Effect, Cause contributes to Effect, or Because of Cause, Effect occurs.* If the paraphrase changes the meaning of the passage, revise or exclude the pair.
- Be careful with detection, measurement, or diagnostic statements. A test may reveal, detect, or identify a condition without causing that condition.
- Prioritise semantic completeness over brevity. Do not shorten spans in a way that removes essential biomedical meaning.

- Treat cue words as prompts for review, not automatic evidence of causality. Words such as *since*, *as*, *therefore*, *thus*, *hence*, and *so* may signal causality in some contexts but may also express time, comparison, explanation, or discourse flow.
- When uncertain, prefer not to annotate unless the row provides enough linguistic evidence to identify a directed cause–effect relation.

E Prompts

We evaluate six prompting strategies across both the extraction and detection tasks. All strategies share the same underlying task definition and causality axes; they differ only in how much reasoning structure and in-context guidance is provided to the model. The strategies are:

- **Zero-Shot** — task definition and rules only, no examples or scaffolding. This serves as the baseline for measuring the model’s prior knowledge of causal extraction.
- **Few-Shot** — adds four labelled input-output examples covering key edge cases (decomposition, hedging, passive voice).
- **Chain-of-Thought (CoT)** — instructs the model to produce explicit step-by-step reasoning before each tuple, encouraging more deliberate modality and cardinality decisions.
- **Hybrid (CoT + Few-Shot)** — combines a worked example with the CoT reasoning format, providing both a process template and a concrete demonstration.
- **Least-to-Most** — decomposes the task into four sequential sub-steps (entity identification, pairing, classification, output), following the least-to-most prompting paradigm.
- **ReAct** — frames the task as a Thought–Action–Observation loop, requiring the model to interleave reasoning and information-gathering actions before committing to a final answer.

E.1 Extraction Prompts

The extraction prompts ask the model to produce structured {Cause, Effect, Causality_Type, Sententiality} tuples for all causal relations in the input. The Zero-Shot prompt below contains the full shared task definition; subsequent prompts omit the repeated system text and show only what is added or changed.

Extraction: Zero-Shot Prompt

System Role: You are an expert causal relation extraction system. Your task is to perform Pairwise Causal Discovery (PCD) on the provided text.

Objective: Extract all causal relations as tuples of {Cause, Effect, Causality_Type, Sententiality}.

1. Definition of Causality. Causation is the “highest form of association.” Distinguish strictly between Cause (agent/trigger) and Effect (outcome). “Can cause” is accepted. Hedged causal claims are not causal facts. Do not extract relations from reported claims (e.g. “Scientists believe X causes Y”). Annotate based on the semantic structure of the text using only common sense; do not decipher abbreviations or apply external definitions.

2. Axis 1 — Marker Presence (Causality Type). *Explicit* ($M=1$): text contains variations of “due to,” “results in,” “leads to,” “causes,” or “effect of.” *Implicit* ($M=0$): causality is present but none of the above markers appear (e.g. “X increased Y,” “X brought about Y”).

Axis 2 — Textual Scope (Sententiality). *Intra-sentential* ($B=0$): Cause and Effect in the same sentence. *Inter-sentential* ($B=1$): Cause and Effect across different sentences.

Axis 3 — Cardinality. Decompose “A causes B and C” into $(A \rightarrow B)$ and $(A \rightarrow C)$ when each cause acts independently. If A and B must combine to cause C, do not decompose.

No transitive closure. If “A causes B, which causes C,” extract $(A \rightarrow B)$ and $(B \rightarrow C)$ only — not $(A \rightarrow C)$ unless explicitly stated.

Passive voice. Canonicalise to active form: “Y is caused by X” → Cause: X, Effect: Y.

Input Text:

[Insert Text Here]

Output:

[Provide extracted tuples here]

Extraction: Few-Shot Prompt

[Same system role, objective, and axes as Zero-Shot above.]

Four examples are added before the task input. They are chosen to cover the most common annotation decisions: relation decomposition (Example 1), hedging exclusion (Example 2), inter-sentential scope and passive voice (Example 3), and joint causation that must not be decomposed (Example 4).

Examples:

Input: “Hypertension leads to heart failure and kidney damage.”

Output:

Tuple 1: {Cause: Hypertension, Effect: heart failure, Causality_Type: Explicit, Sententiality: Intra-sentential}

Tuple 2: {Cause: Hypertension, Effect: kidney damage, Causality_Type: Explicit, Sententiality: Intra-sentential}

Input: “Researchers suggest that excessive sugar intake may cause metabolic syndrome.”

Output: No relations extracted. (*Reason:* “suggest that... may cause” is a reported and hedged claim.)

Input: “Elevated cortisol levels attenuated immune response. This suppression increased susceptibility to infection.”

Output:

Tuple 1: {Cause: Elevated cortisol levels, Effect: immune response attenuation, Causality_Type: Implicit, Sententiality: Intra-sentential}

Tuple 2: {Cause: suppression of immune response, Effect: increased susceptibility to infection, Causality_Type: Implicit, Sententiality: Inter-sentential}

Input: “The combination of alcohol and sedatives can cause respiratory depression.”

Output:

Tuple 1: {Cause: combination of alcohol and sedatives, Effect: respiratory depression, Causality_Type: Explicit, Sententiality: Intra-sentential}

Task — Input Text:

[Insert Text Here]

Output:

Extraction: Chain-of-Thought (CoT) Prompt

[Same system role, objective, and axes as Zero-Shot above.]

The CoT instruction appends a six-step reasoning protocol. Unlike Few-Shot, no labelled examples are provided; instead, the model is asked to produce its own reasoning trace for every candidate relation before committing to a tuple. This is intended to reduce impulsive labelling of hedged or transitive claims.

Instructions. For every potential relationship, perform step-by-step reasoning before outputting a tuple.

1. **Identify Entities** — locate potential Cause and Effect.
2. **Check Modality** — fact or reported claim/hedge? If hedge, discard.
3. **Determine Cardinality** — does the sentence need decomposition?
4. **Determine Causality Type** — check for explicit markers.
5. **Determine Sententiality** — same sentence or across?
6. **Formulate Output** — create the tuple.

Input Text:

[Insert Text Here]

Reasoning & Output:

[Provide step-by-step reasoning then tuples]

Extraction: Hybrid (CoT + Few-Shot) Prompt

[Same system role, objective, and axes as Zero-Shot above.]

This strategy combines a single fully worked example with the CoT reasoning format. The example demonstrates how the reasoning steps map to a real annotation decision — in particular, how passive voice and hedging interact within the same passage.

Example:

Input: “Tumor growth was inhibited by the new compound. However, Dr. Smith claims this might cause fatigue.”

Reasoning:

Sentence 1: Entities — Cause = “new compound,” Effect = “Tumor growth inhibition.” Modality: factual. Passive voice: canonicalise. Causality Type: “inhibited” not in explicit list $\Rightarrow$ Implicit. Sententiality: Intra-sentential.

Sentence 2: “claims... might cause” = reported claim + hedge $\Rightarrow$ rejected.

Final Output:

Tuple 1: {Cause: new compound, Effect: Tumor growth inhibition, Causality_Type: Implicit, Sententiality: Intra-sentential}

Task — Input Text:

[Insert Text Here]

Reasoning:

Final Output:

Extraction: Least-to-Most Prompt

[Same system role, objective, and axes as Zero-Shot above.]

Rather than reasoning within a single pass, the Least-to-Most strategy decomposes extraction into four sequential sub-problems of increasing specificity. This mirrors the original least-to-most prompting framework, where solving easier sub-tasks first provides a scaffold for harder ones such as cardinality decisions and type classification.

Step 1 — Entity Identification. Identify all phrases acting as potential agents (Causes) or outcomes (Effects). Ignore entities in reported claims or hedged statements.

Step 2 — Decomposition & Pairing. Pair identified entities. If a compound cause-effect exists, decide whether they act independently (decompose) or jointly (do not decompose).

Step 3 — Classification. For each valid pair determine: (a) Causality Type — Explicit if “due to / results in / leads to / causes / effect of,” otherwise Implicit; (b) Sententiality — Intra- vs. Inter-sentential.

Step 4 — Final Extraction. Generate the final tuple list from Steps 1–3.

Input Text:

[Insert Text Here]

Extraction: ReAct (Reason + Act) Prompt

[Same system role, objective, and axes as Zero-Shot above.]

The ReAct format frames extraction as an agent loop: each Thought articulates a goal, each Action carries it out on the text, and each Observation records the result. The loop is fixed to four iterations covering scanning, filtering, canonicalisation, and classification — corresponding to the main decision points in the annotation guidelines.

Instruction: Use a Thought–Action–Observation loop.

Thought 1: Scan text for causal markers or semantic causal verbs.

Action 1: Identify candidate sentences.

Observation 1: [List candidate sentences]

Thought 2: Filter by modality (fact vs. claim/hedge).

Action 2: Remove hedged sentences; keep factual causal statements.

Observation 2: [Filtered sentences]

Thought 3: Apply cardinality and passive-voice rules.

Action 3: Decompose independent “AND” lists; convert passive to active.

Observation 3: [Refined pairs]

Thought 4: Classify Axis 1 (Explicit/Implicit) and Axis 2 (Sententiality).

Action 4: Check against explicit marker list; check sentence boundaries.

Final Answer: Output tuples {Cause, Effect, Causality_Type, Sententiality}.

Input Text:

[Insert Text Here]

E.2 Detection Prompts

Detection is a simpler binary task: the model outputs 1 if the passage contains at least one valid causal relation, or 0 otherwise. The same six prompting strategies are applied. The shared causality definition is identical to extraction, but no tuple fields need to be produced — only the existence decision matters. This makes detection a useful diagnostic for whether prompt strategy affects *identification* ability independently of *extraction* precision.

Detection: Zero-Shot Prompt

System Role: You are an expert causal relation detection system. If there is even one causal instance in the text, output 1; if none, output 0.

Definition of Causality (same principles as extraction): strict Cause/Effect distinction; “can cause” is accepted; hedged and reported claims are excluded; annotation based on semantic structure and common sense only.

Input Text:

[Insert Text Here]

Output: “1” or “0”

Detection: Few-Shot Prompt

[Same system role and definition as Detection Zero-Shot above.]

Three examples are provided, chosen to illustrate the two most common failure modes in detection: false negatives from misidentified implicit relations (Example 1 and 3) and false positives from hedged language

that superficially resembles causal claims (Example 2).

Examples:

Input: “Hypertension leads to heart failure and kidney damage.” *Output:* “1”

Input: “Researchers suggest that excessive sugar intake may cause metabolic syndrome.” *Output:* “0”

Input: “Elevated cortisol levels attenuated immune response. This suppression increased susceptibility to infection.” *Output:* “1”

Task — Input Text:

[Insert Text Here]

Output:

Detection: Chain-of-Thought (CoT) Prompt

[Same system role and definition as Detection Zero-Shot above.]

For detection, the CoT protocol is condensed to three steps, since cardinality and sententiality classification are not required. The key reasoning bottleneck is modality: the model must correctly distinguish between factual causal assertions and hedged or attributed claims before issuing a verdict.

Instructions.

1. **Identify Entities** — locate potential Cause and Effect.
2. **Check Modality** — fact or hedge/reported claim? If hedge, discard.
3. **Output** — “1” if any valid causal relation found, “0” otherwise.

Input Text:

[Insert Text Here]

Reasoning & Output:

Detection: Hybrid (CoT + Few-Shot) Prompt

[Same system role and definition as Detection Zero-Shot above.]

The example is chosen to be non-trivial: the passage contains both a valid causal relation and a hedged claim in adjacent sentences. The model must correctly apply modality filtering while recognising that the overall verdict is "1" because at least one valid relation survives after the hedge is removed.

Example:

Input: "Tumor growth was inhibited by the new compound. However, Dr. Smith claims this might cause fatigue."

Reasoning:

Sentence 1 — factual, passive; Cause = "new compound," Effect = "Tumor growth inhibition." Accepted.

Sentence 2 — "claims... might cause" is reported + hedged. Rejected.

At least one valid relation found.

Final Output: "1"

Task — Input Text:

[Insert Text Here]

Reasoning:

Final Output:

Detection: Least-to-Most Prompt

[Same system role and definition as Detection Zero-Shot above.]

For detection, the sub-task decomposition collapses to three steps. Steps 1 and 2 handle identification and filtering; Step 3 aggregates the result into a single binary verdict. This mirrors the extraction variant but stops before classification, since type and sententiality labels are not required.

Step 1 — Entity Identification. Identify all phrases acting as potential agents (Causes) or outcomes (Effects) in the text.

Step 2 — Causal Relation Identification. Ignore entities in reported claims or hedged statements; accept relations per the guidelines.

Step 3 — Final Verdict. Output "1" if at least one valid causal relation was found; "0" if none.

Input Text:

[Insert Text Here]

Detection: ReAct (Reason + Act) Prompt

[Same system role and definition as Detection Zero-Shot above.]

The detection ReAct loop runs for three iterations rather than four, as the canonicalisation step (passive voice, decomposition) is not required to produce a binary output. The loop terminates after validation against the guidelines and issues a verdict.

Instruction: Use a Thought–Action–Observation loop.

Thought 1: Scan for causal markers or semantic causal verbs.

Action 1: Identify candidate sentences.

Observation 1: [List candidate sentences]

Thought 2: Filter by modality.

Action 2: Remove sentences with "believes" or "suggests"; keep factual / "can cause" statements.

Observation 2: [Filtered sentences]

Thought 3: Check remaining sentences against guidelines.

Action 3: Output "1" if any valid causal relation found, "0" otherwise.

Final Answer: [Binary classification]

Input Text:

[Insert Text Here]

F Full Experimental Results

This section provides the complete numeric results for all model– strategy combinations across both tasks. Tables are organised by task (detection then extraction) and within each task by model type (encoder, prompt-only generative, fine-tuned generative, cross-dataset). Reported metrics follow the conventions defined in the main paper.

F.1 Causal Detection — Encoder Models

Table 12 reports the in-distribution detection performance of the four fine-tuned encoder models on the PubMedCausal test split. All four models achieve similar accuracy (> 0.92) and F1 (≈ 0.73), with PubMedBERT leading marginally, likely due to its pre-training on PubMed abstracts closely matching the domain of our corpus.

Table 12: Fine-Tuned Model Performance on Causal Detection.

Model	Acc.	Prec.	Rec.	F1
BERT	0.9284	0.7249	0.7344	0.7296
SciBERT	0.9273	0.7147	0.7440	0.7291
PubMedBERT	0.9304	0.7289	0.7496	0.7391
BioBERT	0.9301	0.7282	0.7481	0.7380

F.2 Causal Detection — Prompt-Only Generative Models

Table 13 reports causal-class precision, recall, and F1 for all eight prompt-only generative models across all six strategies. All metrics are computed on the causal (positive) class only, since the class imbalance makes majority-class accuracy uninformative. Several models collapse to zero recall under structured strategies such as CoT-FewShot and ReAct, suggesting that overly rigid formats cause smaller models to default to the majority (non-causal) class rather than engage with the reasoning scaffold.

F.3 Causal Detection — Fine-Tuned Generative Models

Table 14 shows detection performance after LoRA fine-tuning. Fine-tuning generally improves precision over the prompt-only baselines, but the benefit is uneven across strategies. In particular, Qwen-7B-FT achieves the highest F1 of any generative model (0.3782 under CoT), suggesting that CoT reasoning aligns well with Qwen’s fine-tuned output format. DeepSeek-7B-FT collapses to zero recall under all strategies, indicating that fine-tuning may have overfit the model to a narrow output pattern incompatible with its pre-training inductive biases.

F.4 Causal Extraction — Prompt-Only Generative Models

Table 15 reports extraction performance using three matching levels: Soft (token-overlap F1 on individual Cause and Effect spans and on their pair),

Exact (strict string match on the pair), and Cosine (embedding-similarity match). Soft and Cosine metrics are more lenient and reward semantically correct but lexically inexact extractions; Exact Pair F1 is the strictest signal. Zeros throughout a model’s rows (e.g. DeepSeek-7B) indicate that the model failed to produce parseable output in the expected format under all six strategies.

F.5 Causal Extraction — Fine-Tuned Generative Models

Table 16 reports extraction F1 after LoRA fine-tuning. Mistral-7B-FT with Least-to-Most achieves the strongest overall extraction performance (Soft P-F1 = 0.4245; Cos. P-F1 = 0.5812), suggesting that the structured sub-task decomposition is particularly compatible with Mistral’s instruction-following tendencies after fine-tuning. As with detection, DeepSeek-7B-FT produces zero output under all strategies, confirming a persistent format-alignment failure that fine-tuning did not resolve.

G Cross-Dataset Transfer Detection Setup and Results

This appendix provides the full setup for the cross-dataset causal detection experiment referenced in Section 5.3. The goal of this experiment is to assess whether encoder models trained on PubMedCausal transfer to external causal relation datasets with different domains and annotation schemes. For each dataset, models were first fine-tuned and evaluated on the same dataset to obtain a within-dataset baseline. The cross-dataset setting then evaluated models trained on PubMedCausal directly on the held-out test splits of external datasets without additional fine-tuning on the target dataset.

The transfer results show that PubMedCausal-trained encoders transfer more strongly to BioCause and AltLex than to FinCausal and CTB. The largest positive transfer is observed for BERT on BioCause, where cross-dataset F_1 improves by +0.2027 over the within-dataset baseline. The largest negative transfer is observed for SciBERT on FinCausal, with a drop of -0.2574 . This pattern suggests that transfer is sensitive to domain and annotation-schema alignment: biomedical and general causal datasets benefit more from PubMedCausal training, while financial causal text remains less compatible with the learned decision boundary.

G.1 Cross-Dataset Causal Extraction

Tables 18–19 report extraction performance of base (non-fine-tuned) models on three out-of-distribution datasets. Each table shows Exact, Token-overlap, and Cosine F1 for Cause, Effect, and Pair. The consistent gap between Cosine and Exact F1 across all models and datasets indicates that models frequently recover the correct semantic content but use different surface phrasing than the reference annotation — a known challenge in span-level extraction benchmarks. Tables 21 through 23 show the same breakdown after fine-tuning on PubMedCausal; Llama-8B consistently achieves the strongest generalisation across all three external datasets.

G.2 Error Bucket Definitions

Figure 4 reports the prediction-level error bucketization for the best-performing causal extraction configuration, DeepSeek-R1-32B with few-shot prompting. The analysis is conducted at the cause–effect pair level rather than the sentence level. Each predicted pair is aligned to the closest gold pair within the same input span and assigned to one mutually exclusive bucket using a priority-based scheme. Exact and partial matches are separated from error categories, while the remaining buckets describe the dominant structural failure modes observed in the model outputs.

The distinction between exact and partial matches is important because biomedical causal spans are often long and non-atomic. A prediction may preserve the core causal meaning while differing from the gold annotation in span boundaries. For example, the prediction “higher insulin levels” → “endometrial cancer risk” is not an exact match for the gold pair “high insulin level” → “an increased risk of endometrial cancer”, but it retains the same causal content and is therefore treated as a partial match. The remaining buckets capture genuine extraction failures. Argument-level errors occur when one side of the relation is correct but the other is wrong, as in cause-correct/effect-wrong or effect-correct/cause-wrong cases. Switched relations capture directionality errors, where the model reverses the causal arrow. Spurious predictions indicate unsupported causal pairs generated by the model, while No C-E pair errors indicate annotated gold relations for which the model produced no extraction. This bucketization therefore separates boundary-level variation from more substan-

tive causal extraction failures Table 17.

H Additional Diagnostic Experiments

To better understand model behaviour beyond aggregate extraction scores, we report two compact ablation analyses. First, we compare prompt-only and fine-tuned generative models to test whether LoRA adaptation improves span-level extraction. Also, we separate single-pair and multi-pair causal spans to examine the effect of causal cardinality.

Table 24 shows that LoRA fine-tuning does not consistently improve span-level causal extraction for the two strongest small LLMs. Mistral-7B performs better in its base setting, while Qwen-7B improves on Token Overlap F1 after fine-tuning but does not improve on Exact Pair F1. These results suggest that fine-tuning affects extraction behaviour unevenly.

In Table 26 we conduct a small-LLM causal cardinality ablation by disaggregating the extraction evaluation into single-pair causal spans, multi-pair causal spans and combined causal spans. This ablation isolates the effect of causal relation cardinality on extraction performance.

Table 27 shows that label correctness is highest when extracted spans exactly match the gold spans. Causality classification drops more sharply under partial span matches than sententiality classification, suggesting that Explicit/Implicit labels are more sensitive to span-boundary errors.

Table 13: Causal Detection — Prompt-Only Generative Models (Causal Class Metrics).

Model	Strategy	Accuracy	Precision	Recall	F1
Meta-Llama-3.3-70B	Zero-Shot	0.7568	0.3004	0.6386	0.4086
	Few-Shot	0.4225	0.1584	0.7861	0.2637
	CoT	0.7446	0.2343	0.4151	0.2995
	CoT-FewShot	0.8682	0.1667	0.0005	0.0010
	Least-to-Most	0.4405	0.1796	0.9118	0.3001
	ReAct	0.8680	0.0000	0.0000	0.0000
DeepSeek-R1-Distill-Qwen-32B	Zero-Shot	0.7771	0.1960	0.2240	0.2091
	Few-Shot	0.8201	0.1311	0.0654	0.0873
	CoT	0.5112	0.2000	0.9057	0.3277
	CoT-FewShot	0.8685	0.0000	0.0000	0.0000
	Least-to-Most	0.5921	0.2264	0.8692	0.3592
	ReAct	0.8403	0.0846	0.0218	0.0347
Mixtral-8x7B-Instruct	Zero-Shot	0.6124	0.1989	0.6432	0.3039
	Few-Shot	0.3905	0.1591	0.8479	0.2679
	CoT	0.6310	0.2266	0.7481	0.3478
	CoT-FewShot	0.5535	0.1498	0.5124	0.2319
	Least-to-Most	0.5711	0.2117	0.8302	0.3374
	ReAct	0.8684	0.0000	0.0000	0.0000
Llama-8B	Zero-Shot	0.8089	0.1747	0.1216	0.1434
	Few-Shot	0.7190	0.1273	0.1941	0.1538
	CoT	0.8377	0.2837	0.1536	0.1993
	CoT-FewShot	0.7097	0.2421	0.5666	0.3393
	Least-to-Most	0.8531	0.1379	0.0223	0.0384
	ReAct	0.4702	0.1676	0.7633	0.2748
Mistral-7B	Zero-Shot	0.4247	0.1612	0.8023	0.2684
	Few-Shot	0.1459	0.1317	0.9823	0.2323
	CoT	0.4553	0.1782	0.8697	0.2958
	CoT-FewShot	0.8659	0.1833	0.0056	0.0108
	Least-to-Most	0.8715	0.5799	0.0846	0.1477
	ReAct	0.2907	0.1512	0.9518	0.2609
Llama-3B	Zero-Shot	0.8034	0.1772	0.1358	0.1538
	Few-Shot	0.6576	0.1447	0.3264	0.2005
	CoT	0.8675	0.3171	0.0066	0.0129
	CoT-FewShot	0.5879	0.1825	0.6128	0.2812
	Least-to-Most	0.6993	0.1595	0.3011	0.2085
	ReAct	0.7137	0.1537	0.2610	0.1935
Qwen-7B	Zero-Shot	0.7179	0.2113	0.4187	0.2808
	Few-Shot	0.5771	0.1412	0.4359	0.2133
	CoT	0.6891	0.2122	0.5028	0.2984
	CoT-FewShot	0.7253	0.1775	0.2995	0.2229
	Least-to-Most	0.8587	0.1524	0.0162	0.0293
	ReAct	0.3042	0.1486	0.9067	0.2553
DeepSeek-7B	Zero-Shot	0.3731	0.1597	0.8834	0.2705
	Few-Shot	0.8685	0.0000	0.0000	0.0000
	CoT	0.8685	0.0000	0.0000	0.0000
	CoT-FewShot	0.8685	0.0000	0.0000	0.0000
	Least-to-Most	0.3151	0.1304	0.7420	0.2218
	ReAct	0.1667	0.1351	0.9873	0.2376

Table 14: Causal Detection — Fine-Tuned Generative Models (Causal Class Metrics).

Model	Strategy	Accuracy	Precision	Recall	F1
Qwen-7B-FT	Zero-Shot	0.8528	0.3468	0.1348	0.1942
	Few-Shot	0.7466	0.1700	0.2387	0.1986
	CoT	0.8621	0.4649	0.3188	0.3782
	CoT-FewShot	0.8127	0.2977	0.3122	0.3048
	Least-to-Most	0.8315	0.2934	0.1997	0.2376
	ReAct	0.7995	0.3101	0.4283	0.3597
Llama-3B-FT	Zero-Shot	0.8097	0.1975	0.1460	0.1679
	Few-Shot	0.4756	0.1669	0.7486	0.2730
	CoT	0.8683	0.4969	0.0801	0.1379
	CoT-FewShot	0.6607	0.2094	0.5692	0.3062
	Least-to-Most	0.2846	0.1487	0.9397	0.2568
	ReAct	0.8554	0.3811	0.1591	0.2245
Mistral-7B-FT	Zero-Shot	0.8097	0.1984	0.1470	0.1689
	Few-Shot	0.4752	0.1668	0.7486	0.2729
	CoT	0.8683	0.4953	0.0801	0.1379
	CoT-FewShot	0.6613	0.2095	0.5682	0.3062
	Least-to-Most	0.2842	0.1487	0.9402	0.2568
	ReAct	0.8553	0.3818	0.1612	0.2267
Llama-8B-FT	Zero-Shot	0.8651	0.0820	0.0025	0.0049
	Few-Shot	0.7280	0.1759	0.2899	0.2190
	CoT	0.8628	0.2222	0.0172	0.0320
	CoT-FewShot	0.6564	0.1514	0.3502	0.2114
	Least-to-Most	0.8670	0.0769	0.0010	0.0020
	ReAct	0.8685	0.0000	0.0000	0.0000
DeepSeek-7B-FT	Zero-Shot	0.8677	0.1333	0.0010	0.0020
	Few-Shot	0.8529	0.1047	0.0157	0.0273
	CoT	0.8683	0.2000	0.0005	0.0010
	CoT-FewShot	0.8268	0.1257	0.0532	0.0748
	Least-to-Most	0.7387	0.1350	0.1825	0.1552
	ReAct	0.8685	0.0000	0.0000	0.0000

Table 15: Extraction Performance — Prompt-Only Generative Models (Causal Pair Metrics).

Model	Strategy	Soft C-F1	Soft E-F1	Soft P-F1	Ex. P-F1	Cos. C-F1	Cos. P-F1
DeepSeek-R1-32B	Zero-Shot	0.5366	0.4844	0.5066	0.2222	0.6138	0.5944
	Few-Shot	0.6166	0.5440	0.5758	0.2383	0.7017	0.6765
	CoT	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	CoT-FewShot	0.4297	0.3827	0.4034	0.1811	0.4751	0.4597
	Least-to-Most	0.1114	0.0969	0.1029	0.0456	0.1176	0.1120
	ReAct	0.4749	0.4223	0.4441	0.1775	0.5381	0.5196
Mistral-7B	Zero-Shot	0.4229	0.4293	0.4197	0.1030	0.5551	0.5567
	Few-Shot	0.4743	0.4620	0.4625	0.1428	0.5940	0.5904
	CoT	0.4396	0.4303	0.4292	0.1274	0.5406	0.5380
	CoT-FewShot	0.1809	0.1630	0.1718	0.0658	0.2090	0.2050
	Least-to-Most	0.4990	0.4607	0.4739	0.1527	0.6082	0.5958
	ReAct	0.2207	0.2026	0.2095	0.0889	0.2558	0.2504
Mixtral-8x7B	Zero-Shot	0.3749	0.3355	0.3527	0.1461	0.4240	0.4103
	Few-Shot	0.3922	0.3539	0.3704	0.1469	0.4511	0.4390
	CoT	0.1882	0.1742	0.1812	0.0821	0.2101	0.2063
	CoT-FewShot	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Least-to-Most	0.1356	0.1257	0.1301	0.0658	0.1495	0.1462
	ReAct	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Qwen-7B	Zero-Shot	0.0470	0.0463	0.0463	0.0171	0.0523	0.0525
	Few-Shot	0.3897	0.3464	0.3640	0.1400	0.4429	0.4283
	CoT	0.1966	0.1801	0.1871	0.0703	0.2256	0.2186
	CoT-FewShot	0.1525	0.1362	0.1430	0.0416	0.1774	0.1714
	Least-to-Most	0.1924	0.1734	0.1815	0.0706	0.2166	0.2091
	ReAct	0.2990	0.2798	0.2890	0.1391	0.3302	0.3232
Llama-8B	Zero-Shot	0.1289	0.1186	0.1226	0.0471	0.1484	0.1430
	Few-Shot	0.2848	0.2476	0.2634	0.0911	0.3214	0.3059
	CoT	0.2361	0.2105	0.2200	0.0865	0.2705	0.2578
	CoT-FewShot	0.0450	0.0405	0.0424	0.0186	0.0593	0.0575
	Least-to-Most	0.2216	0.1879	0.2030	0.0874	0.2464	0.2318
	ReAct	0.2952	0.2636	0.2764	0.1089	0.3332	0.3191
Meta-Llama-3.3-70B	Zero-Shot	0.2581	0.2199	0.2384	0.1013	0.2810	0.2676
	Few-Shot	0.0250	0.0217	0.0232	0.0086	0.0280	0.0270
	CoT	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	CoT-FewShot	0.0019	0.0014	0.0016	0.0006	0.0019	0.0017
	Least-to-Most	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	ReAct	0.0006	0.0006	0.0006	0.0006	0.0006	0.0006
Llama-3B	Zero-Shot	0.1494	0.1224	0.1341	0.0526	0.1725	0.1606
	Few-Shot	0.0469	0.0429	0.0436	0.0072	0.0579	0.0550
	CoT	0.1228	0.1129	0.1166	0.0481	0.1388	0.1343
	CoT-FewShot	0.1713	0.1529	0.1606	0.0601	0.1965	0.1885
	Least-to-Most	0.0028	0.0022	0.0025	0.0012	0.0030	0.0027
	ReAct	0.2162	0.2007	0.2059	0.0717	0.2637	0.2554
DeepSeek-7B	Zero-Shot	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Few-Shot	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	CoT	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	CoT-FewShot	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Least-to-Most	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	ReAct	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 16: Extraction Performance — Fine-Tuned Generative Models (Causal Pair Metrics).

Model	Strategy	Soft C-F1	Soft E-F1	Soft P-F1	Ex. P-F1	Cos. C-F1	Cos. P-F1
Mistral-7B-FT	Zero-Shot	0.3224	0.3313	0.3240	0.0756	0.4873	0.4883
	Few-Shot	0.0171	0.0229	0.0199	0.0054	0.0328	0.0345
	CoT	0.3380	0.3601	0.3469	0.0965	0.5001	0.5109
	CoT-FewShot	0.0047	0.0035	0.0040	0.0012	0.0060	0.0056
	Least-to-Most	0.4322	0.4235	0.4245	0.1022	0.5827	0.5812
	ReAct	0.0991	0.1145	0.1059	0.0256	0.2558	0.2594
Qwen-7B-FT	Zero-Shot	0.4137	0.3856	0.3947	0.1293	0.4937	0.4817
	Few-Shot	0.3689	0.3266	0.3439	0.1287	0.4244	0.4091
	CoT	0.1476	0.1312	0.1394	0.0766	0.1612	0.1557
	CoT-FewShot	0.2726	0.2592	0.2645	0.1220	0.3014	0.2962
	Least-to-Most	0.1086	0.0996	0.1041	0.0678	0.1138	0.1112
	ReAct	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Llama-8B-FT	Zero-Shot	0.0216	0.0216	0.0216	0.0080	0.0269	0.0269
	Few-Shot	0.3348	0.3086	0.3175	0.1039	0.4015	0.3890
	CoT	0.2665	0.2530	0.2578	0.0697	0.3263	0.3192
	CoT-FewShot	0.1695	0.1661	0.1666	0.0596	0.2179	0.2159
	Least-to-Most	0.0124	0.0131	0.0126	0.0043	0.0143	0.0142
	ReAct	0.0006	0.0006	0.0006	0.0006	0.0006	0.0006
Llama-3B-FT	Zero-Shot	0.1597	0.1407	0.1477	0.0611	0.1802	0.1731
	Few-Shot	0.1748	0.1496	0.1615	0.0674	0.2002	0.1919
	CoT	0.0568	0.0534	0.0547	0.0204	0.0701	0.0690
	CoT-FewShot	0.0482	0.0436	0.0458	0.0234	0.0607	0.0604
	Least-to-Most	0.0017	0.0014	0.0016	0.0006	0.0019	0.0018
	ReAct	0.0091	0.0083	0.0087	0.0019	0.0110	0.0107
DeepSeek-7B-FT	Zero-Shot	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Few-Shot	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	CoT	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	CoT-FewShot	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Least-to-Most	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	ReAct	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 17: Definitions and examples of prediction-level error bucketization categories. Examples are drawn from DeepSeek-R1-32B Few-shot predictions and the corresponding gold annotations.

Definition	Prediction	Gold
Exact match: The predicted cause and effect spans exactly match the gold cause and effect after normalization.	Prevotellaceae → sepsis	Prevotellaceae → sepsis
Partial match: The prediction is not an exact span match, but it preserves meaningful lexical or semantic overlap with the gold pair under soft-token overlap or cosine similarity.	higher insulin levels → endometrial cancer risk	high insulin level → an increased risk of endometrial cancer
Cause correct, effect wrong: The predicted cause matches the gold cause, but the predicted effect is incorrect, incomplete, or aligned to the wrong outcome.	heterogeneity of both built environment and QoL measures → consistent definitions of both concepts	heterogeneity of both built environment and QoL measures → consistent definitions of both concepts will help advance this area of research
Effect correct, cause wrong: The predicted effect matches the gold effect, but the predicted cause is incorrect or unsupported by the gold annotation.	AIP → AR-HCC	PBGD → AR-HCC
Switched relation: The model reverses the causal direction, such that the predicted cause corresponds to the gold effect and the predicted effect corresponds to the gold cause.	chronic widespread pain → GERD	GERD → chronic widespread pain
Spurious prediction: The model predicts a cause–effect pair that does not reasonably correspond to any gold causal pair in the input.	ADR → diarrhea	Antimicrobials → ADRs
No C-E pair: A gold cause–effect pair exists, but the model produces no causal pair for that sentence or span.	No prediction	the cross-sectional design → causal inference is limited

Table 18: CauseNet Extraction — Base Models.

Model	Metric	Cause F1	Effect F1	Pair F1
Llama-3.2-3B	Exact	0.1689	0.1948	0.0455
	Token	0.1514	0.1584	0.1546
	Cosine	0.3311	0.3224	0.3263
Llama-3.1-8B	Exact	0.1457	0.2333	0.0518
	Token	0.1959	0.2554	0.2265
	Cosine	0.5068	0.5063	0.5065
Mistral-7B	Exact	0.2082	0.1870	0.0593
	Token	0.2726	0.2605	0.2663
	Cosine	0.6222	0.6003	0.6091
Qwen2.5-7B	Exact	0.0153	0.0830	0.0060
	Token	0.0219	0.0436	0.0346
	Cosine	0.0686	0.0722	0.0685

Table 19: BioCause Extraction — Base Models.

Model	Metric	Cause F1	Effect F1	Pair F1
Llama-3.2-3B	Exact	0.0417	0.0774	0.0124
	Token	0.2821	0.1535	0.2219
	Cosine	0.3540	0.2408	0.2998
Llama-3.1-8B	Exact	0.0796	0.0939	0.0254
	Token	0.2847	0.1884	0.2386
	Cosine	0.3438	0.2699	0.3083
Mistral-7B	Exact	0.0485	0.0469	0.0215
	Token	0.1602	0.1553	0.1569
	Cosine	0.2333	0.2335	0.2326
Qwen2.5-7B	Exact	0.0000	0.0351	0.0000
	Token	0.0250	0.0351	0.0305
	Cosine	0.0314	0.0369	0.0342

Table 20: Coling22-Mining Extraction — Base Models.

Model	Metric	Cause F1	Effect F1	Pair F1
Llama-3.2-3B	Exact	0.1858	0.2014	0.0961
	Token	0.2963	0.2802	0.2890
	Cosine	0.3538	0.3406	0.3462
Llama-3.1-8B	Exact	0.3133	0.2901	0.1491
	Token	0.4591	0.4490	0.4518
	Cosine	0.5364	0.5263	0.5282
Mistral-7B	Exact	0.3398	0.2945	0.1739
	Token	0.5718	0.5216	0.5445
	Cosine	0.6803	0.6424	0.6581
Qwen2.5-7B	Exact	0.0535	0.0563	0.0249
	Token	0.0889	0.0845	0.0877
	Cosine	0.1035	0.0896	0.0968

Table 21: CauseNet Dataset — Extraction Performance by Fine-Tuned Model.

Model	Metric	Cause			Effect			Pair		
		P	R	F1	P	R	F1	P	R	F1
Llama-3B	Exact	0.1838	0.1447	0.1619	0.2105	0.2116	0.2111	0.0582	0.0597	0.0589
	Token	0.2183	0.1421	0.1721	0.2566	0.1662	0.2017	0.2374	0.1523	0.1855
	Cosine	0.5887	0.2722	0.3723	0.5340	0.2702	0.3588	0.5614	0.2694	0.3641
Llama-8B	Exact	0.1818	0.2298	0.2030	0.2007	0.3016	0.2410	0.0659	0.0896	0.0759
	Token	0.2846	0.2976	0.2909	0.2789	0.3284	0.3016	0.2818	0.3130	0.2966
	Cosine	0.6268	0.5949	0.6104	0.5971	0.6018	0.5995	0.6120	0.5980	0.6049
Mistral-7B	Exact	0.1872	0.1489	0.1659	0.1139	0.1217	0.1176	0.0323	0.0261	0.0289
	Token	0.2600	0.1841	0.2156	0.1562	0.1186	0.1348	0.2081	0.1514	0.1752
	Cosine	0.5714	0.3661	0.4463	0.5024	0.3272	0.3963	0.5369	0.3439	0.4192
Qwen-7B	Exact	0.2000	0.0255	0.0453	0.2069	0.0635	0.0972	0.0147	0.0037	0.0060
	Token	0.2856	0.0183	0.0343	0.1265	0.0315	0.0504	0.2061	0.0249	0.0444
	Cosine	0.6617	0.0452	0.0847	0.4117	0.0466	0.0838	0.5367	0.0459	0.0846

Table 22: Coling22-Mining Dataset — Extraction Performance by Fine-Tuned Model.

Model	Metric	Cause			Effect			Pair		
		P	R	F1	P	R	F1	P	R	F1
Llama-3B	Exact	0.2756	0.1581	0.2009	0.2564	0.1471	0.1869	0.1183	0.0806	0.0959
	Token	0.4569	0.2201	0.2971	0.4476	0.2244	0.2989	0.4522	0.2198	0.2958
	Cosine	0.5872	0.2510	0.3517	0.5794	0.2552	0.3543	0.5833	0.2505	0.3505
Llama-8B	Exact	0.3250	0.3824	0.3514	0.3047	0.3787	0.3377	0.1506	0.2125	0.1763
	Token	0.5360	0.5743	0.5545	0.5160	0.5876	0.5495	0.5260	0.5783	0.5509
	Cosine	0.6477	0.6539	0.6508	0.6426	0.6711	0.6565	0.6452	0.6600	0.6525
Mistral-7B	Exact	0.2452	0.1397	0.1780	0.1950	0.1140	0.1439	0.0732	0.0440	0.0549
	Token	0.4167	0.2295	0.2960	0.3970	0.2266	0.2885	0.4068	0.2263	0.2908
	Cosine	0.5779	0.3014	0.3961	0.5771	0.3018	0.3963	0.5775	0.2986	0.3937
Qwen-7B	Exact	0.3333	0.0404	0.0721	0.1739	0.0294	0.0503	0.0851	0.0147	0.0250
	Token	0.5305	0.0553	0.1001	0.3055	0.0491	0.0847	0.4180	0.0520	0.0925
	Cosine	0.6480	0.0633	0.1154	0.4541	0.0578	0.1025	0.5510	0.0599	0.1081

Table 23: BioCause Dataset — Extraction Performance by Fine-Tuned Model.

Model	Metric	Cause			Effect			Pair		
		P	R	F1	P	R	F1	P	R	F1
Llama-3B	Exact	0.0087	0.0227	0.0126	0.0413	0.0980	0.0581	0.0000	0.0000	0.0000
	Token	0.2137	0.2097	0.2117	0.1003	0.2538	0.1438	0.1570	0.2268	0.1856
	Cosine	0.2635	0.2742	0.2687	0.1547	0.3102	0.2065	0.2091	0.2907	0.2432
Llama-8B	Exact	0.0130	0.0682	0.0219	0.0290	0.1373	0.0479	0.0079	0.0392	0.0132
	Token	0.1794	0.4029	0.2483	0.0826	0.3656	0.1348	0.1310	0.3778	0.1945
	Cosine	0.2316	0.5759	0.3304	0.1427	0.5345	0.2252	0.1871	0.5465	0.2788
Mistral-7B	Exact	0.0065	0.0227	0.0102	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Token	0.1467	0.2482	0.1844	0.0826	0.2387	0.1228	0.1146	0.2250	0.1519
	Cosine	0.2332	0.3581	0.2825	0.1654	0.3896	0.2322	0.1993	0.3558	0.2555
Qwen-7B	Exact	0.0909	0.0227	0.0364	0.3529	0.1176	0.1765	0.2353	0.0784	0.1176
	Token	0.8464	0.1709	0.2843	0.2538	0.1660	0.2007	0.5501	0.1684	0.2579
	Cosine	0.8757	0.1869	0.3080	0.3957	0.1847	0.2519	0.6357	0.1858	0.2875

Table 24: Fine-tuning ablation for the two strongest small LLMs in causal extraction.

Model	Setting	Token Overlap	Exact	Cos.
Mistral-7B	Base L2M	0.4739	0.1527	0.5958
Mistral-7B	FT L2M	0.4245	0.1022	0.5812
Qwen-7B	Base FS	0.3640	0.1400	0.4283
Qwen-7B	FT ZS	0.3947	0.1293	0.4817

Table 25: Small-LLM cardinality ablation.

Model	Single	Multi	Comb.
Llama-3B	0.0492	0.0131	0.0717
Llama-8B	0.0859	0.0376	0.1089
Mistral-7B	0.1044	0.1777	0.1527
Qwen-7B	0.0762	0.1881	0.1400
DeepSeek-7B	0.0000	0.0000	0.0000

Table 26: Small-LLM causal cardinality ablation. X_{only} contains single-pair causal spans, while Y_{only} contains multi-pair causal spans. The combined row reports the best small-LLM result on the full extraction setting.

Split	Best small-LLM config.	Pair F_1
single-pair	Mistral-7B CoT	0.1044
multiple-pair	Qwen-7B CoT	0.1881
Combined	Mistral-7B L2M	0.1527

Metric	C_{EM}	C_{PM}	C_{AM}	S_{EM}	S_{PM}	S_{AM}
Pairs	818	2,341	3,159	818	2,341	3,159
Accuracy	79.22%	53.14%	59.89%	94.87%	76.80%	81.48%
Macro F1	62.74%	53.10%	58.33%	50.95%	49.18%	50.45%
Weighted F1	80.25%	53.30%	60.66%	96.79%	83.80%	87.33%

Table 27: Model performance for causality and sententiality labeling across exact, partial, and all span-match subsets. **C** = Causality, **S** = Sententiality, **EM** = Exact Match only, **PM** = Partial Match only, and **AM** = All Matches.